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NONPARAMETRIC CLASSIFICATION WITH MISSING DATA

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We introduce a new nonparametric framework for classification problems in the presence of missing data. The key aspect of our framework is that the regression function decomposes into an anova-type sum of orthogonal functions, of which some (or even many) may be zero. Working under a general missingness setting, which allows features to be missing not at random, our main goal is to derive the minimax rate for the excess risk in this problem. In addition to the decomposition property, the rate depends on parameters that control the tail behaviour of the marginal feature distributions, the smoothness of the regression function and a margin condition. The ambient data dimension does not appear in the minimax rate, which can therefore be faster than in the classical nonparametric setting. We further propose a new method, called the *Hard-thresholding Anova Missing data (HAM)* classifier, based on a careful combination of a k-nearest neighbour algorithm and a thresholding step. The HAM classifier attains the minimax rate up to polylogarithmic factors and numerical experiments further illustrate its utility.

1. Introduction. Missing data are ubiquitous in modern statistics, posing a major challenge in a plethora of applications. For instance, missingness may arise when our data are collated from sources that have measured different variables, e.g. in healthcare the data routinely collected on patients may differ across clinics or hospitals. Other reasons include sensor failure, data censoring and privacy concerns, among many others. The topic has been widely studied in many statistical problems; see Josse and Reiter (2018) and Little and Rubin (2019) for a thorough overview. Recent works study missing data in several different contexts, including high-dimensional regression (Loh and Wainwright, 2012; Chandrasekher et al., 2020), covariance and precision matrix estimation (Cai and Zhang, 2016; Loh and Tan, 2018), remote sensing (Zhang et al., 2018), principal component analysis (Elsener and van de Geer, 2019; Zhu et al., 2022), U-statistics (Cannings and Fan, 2022), changepoint detection (Follain et al., 2022) and testing whether the missingness is independent of the data (Berrett and Samworth, 2023; Bordino and Berrett, 2024).

We focus here on missing data in binary classification problems, where the practitioner is presented with the task of assigning a new observation to one of two classes, based on a training set of labelled data. There is a large literature on classification and many algorithms are available in situations when the data is fully observed; see, for example, Devroye et al. (1996) and Boucheron et al. (2005). In settings with missing data, however, there are very few specifically designed classification algorithms, and in many applications practitioners often use ad-hoc approaches to deal with missingness, either resorting to complete-case analysis (i.e. using only the fully observed data points) or removing any covariates for which missing data is present. An alternative option, sometimes used in combination with those mentioned previously, is to impute any missing values to obtain a 'full' data set before employing an existing algorithm. However, the majority of existing classifiers are designed to

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work in idealised settings, where the training and test data arise as fully-observed, independent realisations from the same underlying population; it is often unclear how they perform in combination with these ad-hoc approaches.

The main aim of this paper is to establish a new nonparametric framework for classification problems with missing data. To model the missingness, we will view our training data as independent copies of triples (X, Y, O) arising from a distribution Q on $\mathbb{R}^d \times \{0,1\} \times \{0,1\}^d$, where X denotes a d-dimensional feature vector, Y is its class label and O is the concomitant *observation indicator*. The interpretation here will be that we only observe the components of X for which the corresponding entry of O is 1. The label Y is always observed. For $\omega \in \{0,1\}^d$, we say that (X, Y, O) is an *available case* for ω if the entries of O are 1 whenever the corresponding entry of ω is 1. We write $\eta(x) := \mathbb{P}(Y = 1 | X = x)$ for the *regression function* of Y given X at $x \in \mathbb{R}^d$.

Our framework facilitates the use of all available cases for each observation pattern $\omega \in \{0,1\}^d$. The key observation is that any regression function η on $\mathbb{R}^d$ may be decomposed into an orthogonal sum. That is, we can write

$$\eta(x) = \frac{1}{2} + \sum_{\omega \in \{0,1\}^d} f_\omega(x), \tag{1}$$

where, for each $\omega \in \{0,1\}^d$, the function f_ω in fact only depends on those components of x for which the corresponding entries of ω are 1. We can then estimate the regression function η by recursively estimating each of the f_ω functions starting with $f_{(0,\ldots,0)}$, based on the available cases for ω. One interpretation of (1) is that of an anova decomposition (Efron and Stein, 1981), and the functions f_ω may therefore be viewed as capturing the interactions between the variables. Our framework (see Definition 2) specifies a subset (which may be empty) of observation patterns in $\{0,1\}^d$ for which the corresponding interaction terms are identically 0. The idea here is that the difficulty (in a sense that we make precise later) of the classification problem is determined by the non-zero f_ω, as opposed to the properties of the full regression function in the ambient d-dimensional space.

The primary contribution of the paper is to establish the minimax rate (up to polylogarithmic factors) for the excess risk in this problem (Theorem 1). The rate depends on various parameters that index the class of distributions considered in our framework. Throughout the paper, we will place only mild restrictions on the distribution of the missingness indicator O. Indeed, we neither ask that the components of O are independent of each other, nor that O is necessarily independent of (X, Y), and we even allow for some *missing not at random* settings. See Definition 1 in Section 2 for a precise statement of our conditions on the missingness mechanism. Then, in addition to those imposed by Definitions 1 and 2, we place restrictions on the tail properties of the marginal distribution of the feature vectors (Definition 3), make use of smoothness assumptions on the non-zero f_ω (Definition 4) and employ a standard margin assumption (Definition 5).

Theorem 1 reveals some striking features of our framework. Most notably, we see that consistent (and rate optimal) classification is possible even in situations where we have no complete cases in our training dataset. In contrast, in problems where there is in fact no missing data, we may obtain a faster rate than one could typically expect in a d-dimensional nonparametric classification problem. Overall there is a delicate interplay between the number of available cases for each observation pattern ω and the difficulty of estimating the corresponding function f_ω in our decomposition in (1).

Our next contribution is to propose a new classification algorithm for missing data, introduced formally in Algorithm 1 in Section 3, which we call the *Hard-thresholding Anova Missing data (HAM)* classifier. Our algorithm first estimates the functions f_ω for each $\omega \in \{0,1\}^d$ for which data is available using a nearest neighbour type approach. We then

apply hard-thresholding, setting our final estimator of f_ω to be 0 unless there is sufficient evidence that it is nonzero on the population level. We prove in Theorem 2 that our procedure attains the minimax rate in Theorem 1 up to polylogarithmic factors, subject to a mild sample size condition. We demonstrate the excellent practical performance of our proposal in a numerical study using simulated and real data in Section 4. The key ideas behind the proofs of the upper and lower bounds of our main results are outlined in Section 5. Finally, additional theoretical and numerical results, as well as the full details of the proofs, are presented in the online supplementary material Sell et al. (2023).

Existing work on classification with missing values in the feature vectors includes Josse et al. (2019), who show that in certain settings imputing the missing values with a constant can lead to consistent classification. Cai and Zhang (2019) consider missing data in the setting of high-dimensional linear discriminant analysis; they propose using the available cases to estimate the corresponding class means and variances under a missing completely at random assumption. In regression, Le Morvan et al. (2020) use implicit imputation methods in non-trivial missingness settings for prediction problems, Le Morvan et al. (2021) study a wide class of impute-then-regress procedures, and Ayme et al. (2022) derive minimax rates for prediction using linear models with missing data.

In the related problem of *semi-supervised learning*, the feature vectors are fully observed but some of the class labels in the training data set are missing. This topic has received far more attention, see Ahfock and McLachlan (2023) and Sportisse et al. (2023) for very recent contributions, and the earlier book Chapelle et al. (2006). Similarly, *label noise*, which refers to scenarios where some of the observations in the training dataset have been attributed to the wrong class, has also been widely studied; see the review articles Frénay and Verleysen (2013); Frénay et al. (2014), as well as the recent papers Cannings et al. (2020) and Lee and Barber (2022). Finally, in *transfer learning* (Weiss et al., 2016) the practitioner has access to a small labelled training data set from the target population of interest alongside a larger, noisy source dataset (which may be unlabelled). Here the main interest lies in exploiting the large source sample to improve inferences on the target population; see, for example, Cai and Wei (2021) and Reeve et al. (2021) for recent works in the context of nonparametric classification problems.

It is convenient to now fix some notation that we will use throughout the paper. For $m \in \mathbb{N}$, let $[m] = \{1, \ldots, m\}$. Let $\mathbf{0}_d := (0, \ldots, 0)^T \in \{0,1\}^d$ and $\mathbf{1}_d := (1, \ldots, 1)^T \in \{0,1\}^d$ be the d-dimensional vectors of zeros and ones, respectively. For $j \in [d]$, we write e_j for the jth canonical Euclidean basis vector in $\mathbb{R}^d$, which contains a 1 in the jth component, and 0s elsewhere. For $\omega \in \{0,1\}^d$, we write $d_\omega := \|\omega\|_1$ for the number of 1s in ω, which will refer to as the *dimension* of ω. For $\omega = (\omega_1, \ldots, \omega_d)^T, \omega' = (\omega'_1, \ldots, \omega'_d)^T \in \{0,1\}^d$, we write $\omega \preceq \omega'$ if $\omega_j \leq \omega'_j$ for all $j \in [d]$, $\omega \prec \omega'$ if $\omega_j \leq \omega'_j$ for all $j \in [d]$ and $\omega \neq \omega'$, $\omega \wedge \omega' := (\omega_1 \wedge \omega'_1, \ldots, \omega_d \wedge \omega'_d)^T$, and $\omega \vee \omega' := (\omega_1 \vee \omega'_1, \ldots, \omega_d \vee \omega'_d)^T$. For $x \in \mathbb{R}^d$ and $\omega \in \{0,1\}^d$, we write $x^\omega := x \odot \omega + \mathbf{0}_d \odot (\mathbf{1}_d - \omega) \in \mathbb{R}^d$, and $\odot$ denotes the entrywise product. Further, let $\|\cdot\|$ denote the Euclidean norm and $B_r(x) := \{z \in \mathbb{R}^d : \|z - x\| < r\}$ be the open Euclidean ball of radius $r > 0$ at $x \in \mathbb{R}^d$. For $s \in [d]$, we write $\mathcal{L}_s$ for the Lebesgue measure on $\mathbb{R}^s$. We write $\mathbb{S}^{d \times d}$ for the set of symmetric positive definite matrices. For random variables X, Y and Z we write $X \perp\!\!\!\perp Y$ to denote the independence of X and Y, and $X \perp\!\!\!\perp Y \mid Z$ to denote that X and Y are conditionally independent given Z. Finally, for $x \in \mathbb{R}$, we write $\log_+(x) := \log(x \vee e)$.

2. Statistical setting and minimax results. Let $\mathcal{Q}$ denote the class of all distributions on $\mathbb{R}^d \times \{0,1\} \times \{0,1\}^d$. Then, for $Q \in \mathcal{Q}$, suppose that $(X, Y, O) \sim Q$, and recall that X denotes the feature vector taking values in $\mathbb{R}^d$, Y denotes its class label (either 0 or 1) and O denotes an observation indicator in $\{0,1\}^d$. We write $P \equiv P_Q$ for the marginal distribution of

(X, Y), write $\eta : \mathbb{R}^d \to [0, 1]$ for the regression function given by $\eta(x) := \mathbb{P}_P(Y = 1|X = x)$, and let μ denote the marginal distribution of X on $\mathbb{R}^d$.

For $n \in \mathbb{N}$ and $Q \in \mathcal{Q}$, let $(X_1, Y_1, O_1), \ldots, (X_n, Y_n, O_n)$ be independent and identically distributed triples from Q. In our missing data problem, we only observe those entries of X_i for which the corresponding entry of O_i is 1. In other words, our training data is $D_n := \big((X_1^{O_1}, Y_1, O_1), \ldots, (X_n^{O_n}, Y_n, O_n)\big)$ and we are presented with the task of assigning a fully observed[1] test point X to either class 0 or class 1. Despite replacing the unobserved entries in $X_1^{O_1}, \ldots, X_n^{O_n}$ with zeros, we are still able to distinguish between true (observed) zeros and the zeros that have arisen due to missing values, since $O_1, \ldots, O_n$ are also given in the training data D_n. Recall that, for a pattern $\omega \in \{0, 1\}^d$, we will say that the ith observation is an available case for ω when $\omega \preceq O_i$.

A *data-dependent classifier* $\hat{C}$ is a measurable function from $\mathbb{R}^d \times (\mathbb{R}^d \times \{0, 1\} \times \{0, 1\}^d)^n$ to $\{0, 1\}$, and we let $\mathcal{C}_n$ denote the set of all such data-dependent classifiers. Throughout this paper the latter n arguments of $\hat{C}(x, D_n)$ will always be D_n and thus will often be omitted; we will simply write $\hat{C}(x)$. The performance of $\hat{C} \in \mathcal{C}_n$ will be measured by its *test error*

$$L_P(\hat{C}) := \mathbb{P}_P\big\{\hat{C}(X) \neq Y | D_n\big\} = \int_{\mathbb{R}^d \times \{0,1\}} \mathbb{1}_{\{\hat{C}(x) \neq y\}} \, dP(x, y),$$

which is minimised by the Bayes classifier $C^{\text{Bayes}}(x) = \mathbb{1}_{\{\eta(x) \geq 1/2\}}$. Therefore our main results will concern the (nonnegative) excess test error

$$\mathcal{E}_P(\hat{C}) := L_P(\hat{C}) - L_P(C^{\text{Bayes}}) = \int_{\mathbb{R}^d} \mathbb{1}_{\{\hat{C}(x) \neq C^{\text{Bayes}}(x)\}} |2\eta(x) - 1| \, d\mu(x). \tag{2}$$

We now introduce our new framework for classification problems with missing data. First, in our main results, we will condition on the values of the observation indicators $O_1, \ldots, O_n$ in our training data. We will suppose these belong to a subset $\mathcal{O} \subseteq \{0, 1\}^d$ and one consequence of this is that we only need to consider distributions Q that assign positive mass to each element of $\mathcal{O}$. More formally, we employ the following general assumption on the missingness mechanism.

DEFINITION 1. *For $\mathcal{O} \subseteq \{0, 1\}^d$, let $\mathcal{Q}_{\text{Miss}}(\mathcal{O}) \subseteq \mathcal{Q}$ denote the class of distributions for which $\mathbb{P}_Q(O = o) > 0$ for all $o \in \mathcal{O}$, and*

$$\mathbb{P}_Q(Y = 1|X^\omega = x^\omega) = \mathbb{P}_Q(Y = 1|X^\omega = x^\omega, O = o) \tag{3}$$

for all $x \in \text{supp}(\mu)$, $\omega \in \{0, 1\}^d$ and $o \in \mathcal{O}$ with $\omega \preceq o$.

To understand the generality afforded by Definition 1, first note that (3) holds trivially if O is independent of (X, Y) (i.e. the data is *missing completely at random*) and Q assigns positive probability to $O = o$, for all $o \in \mathcal{O}$, without any additional assumption on the distribution of O nor the distribution P of (X, Y). However, the class $\mathcal{Q}_{\text{Miss}}(\mathcal{O})$ also includes distributions where O and (X, Y) are dependent, and may even allow the data to be *missing not at random*; see the examples below (the proofs of the claims are presented in Section S1.1). Proposition S1 in Section S1.1 provides further understanding of Definition 1 by relating it to conditional independence.

[1] If the test point is not fully observed, then the problem essentially reduces to a lower dimensional classification problem. There will be distributions in our model class where data for the unobserved features in the test point provides no information on the distribution of the observed features.

EXAMPLE 1 ($(1-p)$-homogeneous missingness). *Fix $p \in (0,1)$. Suppose $O \perp\!\!\!\perp (X,Y)$ and the $\mathbb{P}(O=\omega) = p^{d_\omega}(1-p)^{d-d_\omega}$, for $\omega \in \{0,1\}^d$, in other words each component of X is observed (independently) with probability p. Then $Q \in \mathcal{Q}_{\text{Miss}}(\{0,1\}^d)$.*

EXAMPLE 2 (MAR). *Suppose that $\text{supp}(\mu) = [0,1]^4$, and that O is conditionally independent of Y given X^ω for all $\omega \in \{0,1\}^4$. For $x = (x_1,\dots,x_4)^T \in [0,1]^4$, let $\mathbb{P}\{O = (1,1,1,0)^T \mid X = x\} = x_1$ and $\mathbb{P}\{O = (1,0,0,1)^T \mid X = x\} = 1 - x_1$. Then $Q \in \mathcal{Q}_{\text{Miss}}(\{(1,1,1,0)^T, (1,0,0,1)^T\})$.*

EXAMPLE 3 (MNAR). *Let $X \sim U([0,1]^2)$, and for $x = (x_1,x_2)^T \in [0,1]^2$, let $\eta(x) = 1/4 + x_1/2 + 1/4 \cdot \cos(4\pi x_2)$. Suppose also that for $x = (x_1,x_2)^T \in [0,1]^2$ and $r \in \{0,1\}$, we have*

$$\mathbb{P}_Q(O = o \mid X = x, Y = r) = \begin{cases} 1/4 + 1/4 \cdot \mathbb{1}_{\{x_2 \le 1/2\}} & \text{if } o = (1,1)^T \\ 1/6 + 1/12 \cdot \mathbb{1}_{\{x_2 > 1/2\}} & \text{otherwise.} \end{cases}$$

Then $Q \in \mathcal{Q}_{\text{Miss}}(\{0,1\}^2)$.

The aim of these examples is to provide straightforward understanding of how Definition 1 relates to some widely used assumptions in the missing data literature. First, the homogeneous missingness probabilities in Example 1 provide a simple model amenable to theoretical understanding. Though this is at times too restrictive in practical applications, the simplicity allows us to quantify how much the amount of missingness affects the performance in different settings (see, e.g., Loh and Wainwright (2012) and Cai and Zhang (2019)). In Example 2, we will never observe complete cases in our training data set, for instance this situation may arise if different data sources have been merged. As we will see later, this may cause serious problems when trying to apply some of the ad-hoc approaches mentioned in the introduction. Moreover, the data in this example is *Missing At Random* (MAR), in the sense used in, e.g., Josse et al. (2019, Assumption 2). Example 3 covers the challenging setting where data is *Missing Not At Random* (MNAR); the probability that the second component of the 2-dimensional feature vector $(X_1,X_2)^T$ is missing depends on whether or not $X_2 \le 1/2$.

Returning to more general settings, Lemma S2 in the supplementary material shows that (3) holds whenever the distribution of the observation indicator O depends only on a subset of the features and the regression function depends only on another, disjoint subset of the features and these two subsets of features are independent. More precisely, for $S_1, S_2 \subseteq [d]$ with $S_1 \cap S_2 = \emptyset$, suppose the regression function $\eta(x)$ depends on $x = (x_1,\dots,x_d) \in \mathbb{R}^d$ only via $x_{S_1} := \{x_j : j \in S_1\}$ and the conditional distribution of the observation indicator $O|\{X = x, Y = y\}$ depends only on $(x,y) \in \mathbb{R}^d \times \{0,1\}$ via $x_{S_2} := \{x_j : j \in S_2\}$. If X_{S_1} and X_{S_2} are independent then (3) holds.

As outlined in the introduction, a key aspect of our framework is that any regression function $\eta : \mathbb{R}^d \to [0,1]$ may be decomposed into a sum of orthogonal functions $f_\omega : \mathbb{R}^d \to \mathbb{R}$ for $\omega \in \{0,1\}^d$. More precisely, these functions may be defined recursively for $x \in \mathbb{R}^d$ as follows:

$$f_{\mathbf{0}_d}(x) := \mathbb{E}_P\{\eta(X)\} - \frac{1}{2} \equiv \mathbb{P}_P(Y=1) - \frac{1}{2} \tag{4}$$

$$f_\omega(x) := \mathbb{E}_P\Big\{\eta(X) - \frac{1}{2} - \sum_{\omega' \prec \omega} f_{\omega'}(X) \,\Big|\, X^\omega = x^\omega\Big\} \text{ for } \omega \in \{0,1\}^d \text{ with } d_\omega \ge 1. \tag{5}$$

This is known as an *anova decomposition* (Efron and Stein, 1981), and indeed Proposition S3 in Section S1.2 confirms that

$$\eta(\cdot) = \frac{1}{2} + \sum_{\omega \in \{0,1\}^d} f_\omega(\cdot). \tag{6}$$

We will seek to exploit this decomposition to estimate η, by estimating each of the functions $(f_\omega)_{\omega\in\{0,1\}^d}$ in turn. The main idea is that $f_\omega(x)$ only depends on x via x^ω, the entries of x for which the corresponding entries of ω are 1, and thus we may make use of all the available cases for ω when estimating f_ω. Furthermore, if $f_\omega \equiv 0$ for some ω, then we can hope that detecting this will be relatively easy. Indeed, our next definition specifies which of the f_ω functions may contribute to the regression decomposition and which are identically zero.

Before stating Definition 2, it is convenient to introduce some additional notation. For $Q \in \mathcal{Q}_{\mathrm{Miss}}(\mathcal{O})$ and $\omega \in \{0,1\}^d$, we write μ_ω to denote the distribution of X^ω when $(X,Y,O) \sim Q$, i.e. the distribution on $\mathbb{R}^d$ given by $\mu_\omega(A) = \mu\big(\{x \in \mathbb{R}^d : x^\omega \in A\}\big)$ for measurable $A \subseteq \mathbb{R}^d$. Moreover, for $o \in \mathcal{O}$ and $\omega \preceq o$, we write $\mu_{\omega|o}$ to denote the distribution of $X^\omega \mid \{O = o\}$ when $(X,Y,O) \sim Q$, i.e. the distribution on $\mathbb{R}^d$ given by $\mu_{\omega|o}(A) = \mathbb{P}_Q(X^\omega \in A \mid O = o)$, for measurable $A \subseteq \mathbb{R}^d$. For $Q \in \mathcal{Q}_{\mathrm{Miss}}(\mathcal{O})$ and $\omega \in \{0,1\}^d$, define

$$\sigma^2_\omega := \min_{o\in\mathcal{O}:o\succeq\omega} \mathbb{E}_Q\big\{f^2_\omega(X) \,\big|\, O = o\big\}. \tag{7}$$

To make a precise statement of our main condition, which asks that f_ω is zero for certain values of ω, we will also make use of some basic notation concerning orderings of subsets of $\{0,1\}^d$. Let $\mathcal{I}(\{0,1\}^d) := \{\Omega \subset \{0,1\}^d : \text{if } \omega,\omega' \in \Omega \text{ and } \omega \preceq \omega', \text{ then } \omega = \omega'\}$ be the set of *antichains* in $\{0,1\}^d$, i.e. the set of subsets $\Omega \subseteq \{0,1\}^d$ for which any two distinct elements of Ω are *incomparable*. Further, for $\Omega \in \mathcal{I}(\{0,1\}^d)$, let $L(\Omega) := \{\omega \in \{0,1\}^d : \text{ there exists } \omega' \in \Omega \text{ such that } \omega \prec \omega'\}$ denote the elements of $\{0,1\}^d$ that precede the elements of Ω, and write $U(\Omega) := \{0,1\}^d \setminus \{\Omega \cup L(\Omega)\}$.

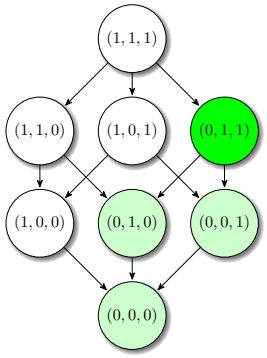


FIG 1. *A Hasse diagram representing the role of $\Omega_\star$ used in Definition 2. In this example, $d = 3$ and $\Omega_\star = \{(0,1,1)^T\}$ (in dark green), we present $L(\Omega_\star) = \{(0,1,0)^T, (0,0,1)^T, \mathbf{0}_d\}$ in light green and $U(\Omega_\star)$ in white.*

DEFINITION 2. *For $\Omega_\star \in \mathcal{I}(\{0,1\}^d)$, $c_{\mathrm{E}} \in [0,1/4]$ and $\mathcal{O} \subseteq \{0,1\}^d$, let*

$$\mathcal{Q}_{\mathrm{E}}(\Omega_\star, c_{\mathrm{E}}, \mathcal{O}) := \Big\{Q \in \mathcal{Q}_{\mathrm{Miss}}(\mathcal{O}) : \sigma^2_\omega \geq c_{\mathrm{E}} \text{ for all } \omega \in \Omega_\star,\ f_\omega \equiv 0 \text{ for all } \omega \in U(\Omega_\star)\Big\}.$$

Definition 2 is very flexible. For example, if $\Omega_\star = \big\{\omega \in \{0,1\}^d : d_\omega = 1\big\}$, then it asks that η is an *additive* function of the components of x (see, for example, Hastie and Tibshirani (1986)). As another example, suppose $S \subseteq [d]$ and $\Omega_\star = \{(\omega_1,\ldots,\omega_d)^T \in \{0,1\}^d : \omega_j = 0 \text{ for } j \in S^c\}$, then η only depends on the components of x that are contained in S. Other, trivial, cases include $\Omega_\star = \{\mathbf{1}_d\}$, which (when $c_{\mathrm{E}} = 0$) corresponds to no assumption, $\Omega_\star = \{\mathbf{0}_d\}$ asks that η is constant, and $\Omega_\star = \emptyset$ forces $\eta \equiv 1/2$. Further examples are given in Section 4, based on specific instances of Examples 1 to 3.

The condition that $\sigma_\omega^2 \geq c_{\mathrm{E}}$ for $\omega \in \Omega_\star$ is redundant if c_{E} is zero, and we may indeed take $c_{\mathrm{E}} = 0$ in Theorem 1 below. The main role of the constant c_{E} is in Theorem 2, where it is assumed to be strictly positive. The intuition is that this provides a natural measure of the signal strength, and that, with enough data, we are able to detect which patterns ω belong to $\Omega_\star$. The minimum in the definition of σ_ω^2 accounts for the fact that we will ultimately use the available cases in our training data to estimate the signal strength from each ω, and that our results will be presented conditional on $O_1, \ldots, O_n$.

Our next condition concerns the regularity of the marginal feature distributions. For a probability distribution ν on $\mathbb{R}^d$ and $s \in [0, d]$ let $\rho_{\nu,s} : \mathbb{R}^d \to [0, 1]$ denote the *lower density* (Reeve et al., 2021) given by

$$\rho_{\nu,s}(x) := \inf_{r \in (0,1)} \frac{\nu\big(B_r(x)\big)}{r^s}. \tag{8}$$

The term 'lower density' is used to highlight the relationship between $\rho_{\nu,d}$ and the density (if it exists) of ν with respect to Lebesgue measure on $\mathbb{R}^d$. Note, however, that our restriction on the marginal feature distributions in Definition 3 below does not require that these marginal distributions have a density, nor that the lower density is uniformly bounded below by a positive constant.

DEFINITION 3. *Fix $\boldsymbol{\gamma} = (\gamma_1, \ldots, \gamma_d)^T \in [0, \infty)^d$, $C_{\mathrm{L}} \geq 1$ and $\mathcal{O} \subseteq \{0, 1\}^d$. For $\omega = (\omega_1, \ldots, \omega_d)^T \in \{0, 1\}^d \setminus \{\mathbf{0}_d\}$, let $\gamma_\omega := \min\{\gamma_j : \omega_j = 1\}$. Let $\mathcal{Q}_{\mathrm{L}}(\boldsymbol{\gamma}, C_{\mathrm{L}}, \mathcal{O})$ denote the subclass of $\mathcal{Q}_{\mathrm{Miss}}(\mathcal{O})$ for which*

$$\mu_\omega\Big(\Big\{x \in \mathbb{R}^d : \min_{o \in \mathcal{O} : \omega \preceq o} \rho_{\mu_{\omega|o}, d_\omega}(x) < \xi\Big\}\Big) \leq C_{\mathrm{L}} \cdot \xi^{\gamma_\omega} \tag{9}$$

for all $\xi > 0$ and all $\omega \in \{0, 1\}^d \setminus \{\mathbf{0}_d\}$.

To put Definition 3 into context, first suppose that O is independent of (X, Y), so that $\mu_{\omega|o} = \mu_\omega$ for all $o \in \mathcal{O}$ and thus $\min_{o \in \mathcal{O} : \omega \preceq o} \rho_{\mu_{\omega|o}, d_\omega}(\cdot) = \rho_{\mu_\omega, d_\omega}(\cdot)$. Then the parameters γ_j for $j \in [d]$ control the tail properties of the univariate marginal distributions of μ, where in particular a smaller value of γ_j allows the tails to be heavier. Moreover, the definition of γ_ω ensures that $\gamma_\omega \geq \gamma_{\omega'}$ whenever $\omega \preceq \omega'$. Thus for $Q \in \mathcal{Q}_{\mathrm{L}}(\gamma, C_{\mathrm{L}}, \mathcal{O})$, the tails of the corresponding marginal distribution $\mu_{\omega'}$ are heavier compared with the μ_ω one. This is a natural restriction, concatenating a heavy tailed variable with a light tailed one shouldn't result in a light tailed joint distribution. Indeed, Reeve et al. (2021, Proposition S8 and Lemma S9) show that this property holds for product measures; in fact that paper provides many additional appealing aspects of lower densities. As an illustration, suppose that $(X, Y, O) \sim Q \in \mathcal{Q}_{\mathrm{Miss}}(\mathcal{O})$, where $X \sim U([0, 1]) \otimes N(0, 1)$, and O is independent of (X, Y). Then we may take any $\gamma_1 \in [0, \infty)$ and $\gamma_2 \in [0, 1]$ for the univariate marginals (see Section S1.1). Then, letting $\boldsymbol{\gamma} = (\gamma_1, \gamma_2)$, Reeve et al. (2021, Proposition S8) confirms that $Q \in \mathcal{Q}_{\mathrm{L}}(\boldsymbol{\gamma}, C_{\mathrm{L}}, \mathcal{O})$, for sufficiently large C_{L} (depending on $\boldsymbol{\gamma}$).

In settings where the missingness O depends on (X, Y), the minimum over $o \in \mathcal{O}$ in (9) provides control of the measure of the distributions of $X^\omega | \{O = o\}$ from which we observe training data. Note that the measure on the outside in (9) is μ_ω, the marginal distribution of X^ω of our fully observed test point X. Intuitively, (9) controls the relationship between the distributions of X^ω and $X^\omega | \{O = o\}$ for $o \in \mathcal{O}$. The case $\omega = \mathbf{0}_d$ is excluded from Definition 3, since each one of the measures $\mu_{\mathbf{0}_d}$ and $\mu_{\mathbf{0}_d|o}$ for $o \in \{0, 1\}^d$ are the point mass at zero, and thus (9) holds with any $\gamma_{\mathbf{0}_d} \geq 0$ and $C_{\mathrm{L}} = 1$.

In Section S1.1, we study specific instances of our examples above and in particular characterise which tail parameters we can choose in the class $\mathcal{Q}_{\mathrm{L}}(\boldsymbol{\gamma}, C_{\mathrm{L}}, \mathcal{O})$. In general, for compactly supported variables with lower densities bounded away from zero the tail parameter

can be taken arbitrarily large, and for variables supported on all of $\mathbb{R}$ with infinitely many moments the tail parameter may be taken to be any value smaller than one.

The remaining conditions on our class of distributions concern the smoothness of the functions f_ω, which contribute to the decomposition in (6), and a standard margin assumption on the regression function η. These two conditions only concern the marginal distribution of the pair (X, Y), and are therefore presented as subsets of the class $\mathcal{P}$ of all distributions on $\mathbb{R}^d \times \{0,1\}$.

DEFINITION 4. *Fix $\boldsymbol{\beta} = (\beta_1, \ldots, \beta_d)^T \in (0,1]^d$ and $C_{\mathrm{S}} \geq 1$. For $\omega = (\omega_1, \ldots, \omega_d)^T \in \{0,1\}^d \setminus \{\mathbf{0}_d\}$, let $\beta_\omega := \min\{\beta_j : \omega_j = 1\}$. Let $\mathcal{P}_{\mathrm{S}}(\boldsymbol{\beta}, C_{\mathrm{S}})$ denote the class of distributions for which*

$$|f_\omega(x_1) - f_\omega(x_2)| \leq C_{\mathrm{S}} \cdot \|x_1^\omega - x_2^\omega\|_2^{\beta_\omega} \tag{10}$$

for all $x_1, x_2 \in \mathrm{supp}(\mu)$ and all $\omega \in \{0,1\}^d \setminus \{\mathbf{0}_d\}$.

The form of β_ω in Definition 4 imposes a mild natural ordering on the smoothness, namely that $f_{\omega'}$ is at least as smooth as f_ω when $\omega' \prec \omega$. This only rules out situations where higher-order interactions in the anova decomposition of η in (6) are allowed to be smoother than the corresponding main univariate contributions.

In our main results below, we will consider distributions Q in the intersection of $\mathcal{Q}_{\mathrm{E}}(\Omega_\star, c_{\mathrm{E}}, \mathcal{O})$ from Definition 2 and for which the corresponding P belongs to $\mathcal{P}_{\mathrm{S}}(\boldsymbol{\beta}, C_{\mathrm{S}})$, thus (10) is in fact only imposing an additional constraint for $\omega \in \Omega_\star \cup L(\Omega_\star)$. A straightforward calculation shows that if $Q \in \mathcal{Q}_{\mathrm{E}}(\Omega_\star, c_{\mathrm{E}}, \mathcal{O})$ and $P \equiv P_Q \in \mathcal{P}_{\mathrm{S}}(\boldsymbol{\beta}, C_{\mathrm{S}})$, then the corresponding regression function satisfies

$$|\eta(x_1) - \eta(x_2)| \leq 2^d \cdot C_{\mathrm{S}} \cdot \|x_1 - x_2\|_2^{\min_{\omega \in \Omega_\star}\{\beta_\omega\}}$$

for all $x_1, x_2 \in \mathrm{supp}(\mu)$.

Our final definition is the standard margin condition (e.g. Polonik, 1995; Mammen and Tsybakov, 1999).

DEFINITION 5. *Fix $\alpha \in [0, \infty)$ and $C_{\mathrm{M}} \geq 1$. Let $\mathcal{P}_{\mathrm{M}}(\alpha, C_{\mathrm{M}})$ denote the class of distributions for which*

$$\mu\Big(\big\{x \in \mathbb{R}^d : \big|\eta(x) - 1/2\big| < t\big\}\Big) \leq C_{\mathrm{M}} \cdot t^\alpha$$

for all $t > 0$.

In Theorem 1 below we provide our main minimax result, in which the class of distributions of interest is the intersection of the classes given in Definitions 1 to 5. More precisely, for $\Omega_\star \in \mathcal{I}(\{0,1\}^d) \setminus \{\{\mathbf{0}_d\}, \emptyset\}$, $c_{\mathrm{E}} \in [0, 1/4]$, $\boldsymbol{\gamma} \in [0, \infty)^d$, $C_{\mathrm{L}} > 1$, $\boldsymbol{\beta} \in (0,1]^d$, $C_{\mathrm{S}} \geq 1$, $\alpha \in [0, \infty)$, $C_{\mathrm{M}} \geq 1$ and $\mathcal{O} \subseteq \{0,1\}^d$, we write

$$\begin{aligned} \mathcal{Q}'_{\mathrm{Miss}} &\equiv \mathcal{Q}'_{\mathrm{Miss}}(\Omega_\star, c_{\mathrm{E}}, \boldsymbol{\gamma}, C_{\mathrm{L}}, \boldsymbol{\beta}, C_{\mathrm{S}}, \alpha, C_{\mathrm{M}}, \mathcal{O}) \\ &:= \mathcal{Q}_{\mathrm{E}}(\Omega_\star, c_{\mathrm{E}}, \mathcal{O}) \cap \mathcal{Q}_{\mathrm{L}}(\boldsymbol{\gamma}, C_{\mathrm{L}}, \mathcal{O}) \cap \{Q \in \mathcal{Q} : P \equiv P_Q \in \mathcal{P}_{\mathrm{S}}(\boldsymbol{\beta}, C_{\mathrm{S}}) \cap \mathcal{P}_{\mathrm{M}}(\alpha, C_{\mathrm{M}})\}. \end{aligned} \tag{11}$$

As mentioned above, in the theorem we consider the excess risk conditional on the values of the observation indicators $O_1, \ldots, O_n$. To that end, for fixed $o_1, \ldots, o_n \in \mathcal{O}$, let $n_\omega := \sum_{i=1}^n \mathbb{1}_{\{\omega \preceq o_i\}}$ denote the number of available cases for the observation pattern $\omega \in \{0,1\}^d$, and define $\mathcal{N} := \{\omega \in \{0,1\}^d : n_\omega > 0\}$ to be the set of observation patterns for which we have training data.

THEOREM 1. *Fix* $d, n \in \mathbb{N}$, $\mathcal{O} \subseteq \{0,1\}^d$, $o_1, \ldots, o_n \in \mathcal{O}$, $\Omega_\star \in \mathcal{I}(\{0,1\}^d) \setminus \{\{\mathbf{0}_d\}, \emptyset\}$, $c_{\mathrm{E}} \in [0, 1/4]$, $\boldsymbol{\gamma} \in [0,\infty)^d$, $C_{\mathrm{L}} > 1$, $\boldsymbol{\beta} \in (0,1]^d$, $C_{\mathrm{S}} \geq 1$, $\alpha \in [0,\infty)$ *and* $C_{\mathrm{M}} \geq 1$. *Suppose that* $\min_{\omega \in \Omega_\star} \gamma_\omega > \max_{\omega \in \Omega_\star} \beta_\omega / d_\omega$, $\max_{\omega \in \Omega_\star} \alpha\beta_\omega \leq \min_{\omega \in \Omega_\star} d_\omega$, $c_{\mathrm{E}} \leq (16|\Omega_\star|)^{-1}$, $C_{\mathrm{L}} \geq (8d)^{(1+d)(1+\|\boldsymbol{\gamma}\|_\infty)}$ *and* $C_{\mathrm{M}} \geq \max_{\omega \in \Omega_\star} \left(1 + 6 \cdot 4^{d/\beta_\omega}\right)$. *Let*

$$R := \max_{\omega \in \Omega_\star \cap \mathcal{N}} \left\{ n_\omega^{-\frac{\beta_\omega \gamma_\omega (1+\alpha)}{\gamma_\omega (2\beta_\omega + d_\omega) + \alpha\beta_\omega}} \right\}. \tag{12}$$

Then, there exist constants $0 < c < C$ *(neither of which depend on* n *nor* $o_1, \ldots, o_n$*) such that*

$$\begin{aligned} c \cdot \left(R + \mathbb{1}_{\{\Omega_\star \cap \mathcal{N}^{\mathrm{c}} \neq \emptyset\}} \right) &\leq \inf_{\hat{C} \in \mathcal{C}_n} \sup_{Q \in \mathcal{Q}'_{\mathrm{Miss}}} \mathbb{E}_Q \left\{ \mathcal{E}_{P_Q}(\hat{C}) \,\middle|\, O_1 = o_1, \ldots, O_n = o_n \right\} \\ &\leq C \cdot \log_+^{\frac{1+\alpha}{2}} \left(\min_{\omega \in \Omega_\star \cap \mathcal{N}} n_\omega \right) \cdot R + \mathbb{1}_{\{\Omega_\star \cap \mathcal{N}^{\mathrm{c}} \neq \emptyset\}}. \end{aligned} \tag{13}$$

The proof of Theorem 1 is given in Section 5. The upper bound in (13) is attained by an *oracle* version of our algorithm presented in Section 3, which is allowed access to the set $\Omega_\star$. Note, however, that our HAM classifier in Algorithm 1, which doesn't have access to $\Omega_\star$ also attains the same upper bound (up to poly-logarithmic factors), subject to a sample size condition; see Theorem 2 for a precise statement.

Theorem 1 establishes the minimax rate of convergence (up to polylogarithmic factors) under our framework. In settings where $\Omega_\star \subseteq \mathcal{N}$, the rate is given by the quantity R in (12), which elucidates the delicate interplay between the parameters that index our class $\mathcal{Q}'_{\mathrm{Miss}}$, as well as the number of available cases for each of the observation patterns in $\Omega_\star$. To provide some further intuition, the quantity $n_\omega^{-\frac{\beta_\omega \gamma_\omega (1+\alpha)}{\gamma_\omega (2\beta_\omega + d_\omega) + \alpha\beta_\omega}}$ reflects the contribution to the rate arising from the difficulty of estimating a non-zero ω-way interaction term f_ω in the anova decomposition in (1). Crucially, only the n_ω available cases in the training data set may be used to estimate this interaction term. Then the rate R we obtain is given by the worst case over those ω for which the corresponding f_ω is non-zero as long as we have data available with which to estimate these, i.e. $\Omega_\star \subseteq \mathcal{N}$. Given two observation patterns $\omega \succ \omega' \in \{0,1\}^d$, we have $n_\omega \leq n_{\omega'}$, $\beta_\omega \leq \beta_{\omega'}$, $\gamma_\omega \leq \gamma_{\omega'}$, and $d_\omega > d_{\omega'}$. Therefore we have

$$n_\omega^{-\frac{\beta_\omega \gamma_\omega (1+\alpha)}{\gamma_\omega (2\beta_\omega + d_\omega) + \alpha\beta_\omega}} > n_{\omega'}^{-\frac{\beta_{\omega'} \gamma_{\omega'} (1+\alpha)}{\gamma_{\omega'} (2\beta_{\omega'} + d_{\omega'}) + \alpha\beta_{\omega'}}}.$$

The consequence of this is that if f_ω is non-zero, then estimating $f_{\omega'}$ will be easier than estimating f_ω. This is why the maximum in R is taken only over the observation patterns in $\Omega_\star$. When $|\Omega_\star| \geq 2$, the maximum in R may be attained by a pattern ω with large values of γ_ω and β_ω, and low dimension d_ω, but for which we only have few available cases. Alternatively, it may be that the rate is driven by a pattern ω for which a large quantity of data is available, but the corresponding f_ω is difficult to estimate (i.e. with unfavourable γ_ω, β_ω, and d_ω). We provide explicit examples of the rate R that we encounter in three different settings in Section 4.

The indicator function on the left-hand side in (13) shows that if there are no available cases for one of the patterns in $\Omega_\star$, then the worst case excess risk for any algorithm is bounded below by a constant $c > 0$. The intuition is that, for the distributions in our class where the ω-way interaction term f_ω is non-zero, this interaction may only be accurately estimated using observations that are available cases for ω. The corresponding upper bound in this case follows from the fact that the excess risk is always bounded by 1.

Multiple strengths of our framework are highlighted by Theorem 1. We see that if $\mathbf{1}_d \notin \Omega_\star$, then consistent (and rate optimal) classification is still possible even when we have no complete cases in our training data set. Moreover, when there is actually no missing data (so

$n_\omega = n$ for every ω), the rate under our framework is $n^{-\min_{\omega\in\Omega_\star}\frac{\beta_\omega\gamma_\omega(1+\alpha)}{\gamma_\omega(2\beta_\omega+d_\omega)+\alpha\beta_\omega}}$. Comparing this rate to what one may expect when $\Omega_\star = \{\mathbf{1}_d\}$ and $c_{\mathrm{E}} = 0$ (thus placing no restriction in Definition 2), we see that we obtain a faster rate of convergence than in the corresponding d-dimensional classification problem under just tail, smoothness and margin conditions, since $\min_{\omega\in\Omega_\star}\frac{\beta_\omega\gamma_\omega(1+\alpha)}{\gamma_\omega(2\beta_\omega+d_\omega)+\alpha\beta_\omega} \geq \frac{\beta_{\min}\gamma_{\min}(1+\alpha)}{\gamma_{\min}(2\beta_{\min}+d)+\alpha\beta_{\min}}$, where $\beta_{\min} := \beta_{\mathbf{1}_d}$. Here the inequality is strict if $\mathbf{1}_d \notin \Omega_\star$. Finally, in situations where all of the variables are important but there are limited interactions between the covariates, for instance in an additive model with $\Omega_\star = \{\omega \in \{0,1\}^d : d_\omega = 1\}$, our rate R does not depend on the ambient dimension d.

In the theorem, asking that $\Omega_\star \notin \{\{\mathbf{0}_d\}, \emptyset\}$ is only ruling out trivial cases. When $\Omega_\star = \{\mathbf{0}_d\}$, the regression functions are constant and the classification problem reduces to a binomial testing problem, where based on just the labels $Y_1, \ldots, Y_n \sim \mathrm{Bernoulli}(1/2 + f_{\mathbf{0}_d})$ we would like to test whether $f_{\mathbf{0}_d}$ is greater or less than 0. If $\Omega_\star = \emptyset$, the regression function is the constant $1/2$, and thus we can obtain an excess risk of 0 with a random guess (in fact, the excess risk is 0 for any classifier). The restrictions on the relationships between the parameters are also mild. The condition that $\min_{\omega\in\Omega_\star}\gamma_\omega > \max_{\omega\in\Omega_\star}\beta_\omega/d_\omega$ (which is only needed for the upper bound) rules out distributions where the tails of the marginal distributions are so heavy that there is very little training data in regions where there is a reasonable chance that we may observe our test point. The requirement that $\max_{\omega\in\Omega_\star}\alpha\beta_\omega \leq \min_{\omega\in\Omega_\star} d_\omega$ (which is only needed for the lower bound), is related to the condition that $\alpha\beta \leq d$ used in the standard d-dimensional nonparametric classification problem. This latter condition rules out *super-fast rates* (Audibert and Tsybakov, 2007), and it holds under mild conditions (see Reeve et al. (2021, Lemma S15)). The additional constraints on the constants c_{E}, C_{L} and C_{M} are also only used in the proof of the lower bound; these restrictions may be relaxed slightly (indeed the bounds used in the proofs in Section 5.1 are weaker), but for brevity we present the simpler bounds in the statement of the theorem.

Finally, note that throughout this section, and in fact the whole paper, we suppose that our test point follows the marginal distribution $P \equiv P_Q$ of (X, Y) when $(X, Y, O) \sim Q$. One may be interested instead in settings where the test point is arising from the conditional distribution $(X, Y)|\{O = \mathbf{1}_d\}$, say; in this case our theory may still be applied with minor changes, e.g., μ in (2) (and other places) should be replaced with $\mu_{\mathbf{1}_d|\mathbf{1}_d}$.

3. The Hard-thresholding Anova Missing data classifier . In this section we introduce our new algorithm for classification with missing data. As mentioned in the previous section, we do not need to know in advance which of the interaction terms in (6) are nonzero. The full procedure, which we call the Hard-thresholding Anova Missing data (HAM) classifier, is given in Algorithm 1. The main result in this section (Theorem 2) establishes that the HAM classifier achieves the upper bound in Theorem 1, subject to a sample size condition, despite the fact that our algorithm does not have access to the set $\Omega_\star$ from Definition 2.

Before presenting our theoretical results, we describe the key ideas underpinning our approach. The first main step begins by estimating $\mathbb{P}(Y = 1)$ via the proportion of labels in the training data that belong to class 1, which provides an estimate of $f_{\mathbf{0}_d}$. The algorithm then proceeds to estimate each of the functions f_ω for $\omega \in \mathcal{N}$, in order of increasing dimension d_ω. More precisely, for each $\omega \in \mathcal{N}$, the estimator $\hat{f}_\omega$ of f_ω is based on a nearest neighbour type algorithm, which uses only the available cases for ω, and where the distance metric is based on the covariates for which the corresponding entries of ω are equal to 1.

The second main step of the algorithm aims to determine which of the functions f_ω should be set to zero. We first calculate a proxy for σ^2_ω given in (7) by taking the average after evaluating the estimator $\hat{f}^2_\omega$ at each of the available cases for ω in the training dataset. We then seek to estimate $\Omega_\star$. Starting with $\omega = \mathbf{1}_d$, and now working in order of decreasing dimension

Algorithm 1 The Hard-thresholding Anova Missing data classifier $\hat{C}_{\mathrm{HAM}}$.

Input: Data $D_n = ((X_1^{o_1}, Y_1, o_1), \ldots, (X_n^{o_n}, Y_n, o_n)) \in (\mathbb{R}^d \times \{0,1\} \times \{0,1\}^d)^n$, $\boldsymbol{\gamma} \in [0,\infty)^d$, $\boldsymbol{\beta} \in (0,1]^d$, $\alpha \in [0,\infty)$, and a test point $x_0 \in \mathbb{R}^d$
$\hat{f}_{\mathbf{0}}(\cdot) := \frac{1}{n}\sum_{i=1}^n Y_i - \frac{1}{2}$
for $d' = 1, \ldots, d$ **do**
  for $\omega \in \mathcal{N}$ such that $d_\omega = d'$ **do**
    $k_\omega := 1 + \lfloor n_\omega^{\frac{2\beta_\omega\gamma_\omega}{\gamma_\omega(2\beta_\omega+d_\omega)+\alpha\beta_\omega}} \rfloor$; $\tau_\omega := 2^{-4} \cdot n_\omega^{-\frac{\beta_\omega\gamma_\omega}{2[\gamma_\omega(2\beta_\omega+d_\omega)+\alpha\beta_\omega]}}$; $N_\omega := \{i \in [n] : \omega \preceq o_i\}$
    for $x \in \{X_i^{o_i} : i \in N_\omega\} \cup \{x_0\}$ **do**
      Let $(X^\omega_{(1)_\omega}(x), Y_{(1)_\omega}(x)), \ldots, (X^\omega_{(n_\omega)}(x), Y_{(n_\omega)_\omega}(x))$ be a reordering of the pairs $\{(X_i^{o_i}, Y_i) : i \in N_\omega\}$ such that $\|X^\omega_{(1)_\omega}(x) - x^\omega\| \leq \cdots \leq \|X^\omega_{(n_\omega)_\omega}(x) - x^\omega\|$
      $\hat{f}_\omega(x) := \frac{1}{k_\omega}\sum_{j=1}^{k_\omega} Y_{(j)_\omega}(x) - \frac{1}{2} - \sum_{\omega' \prec \omega} \hat{f}_{\omega'}(x)$
    end for
    $\hat{\sigma}^2_\omega := \frac{1}{n_\omega}\sum_{i \in N_\omega} \hat{f}^2_\omega(X_i^\omega)$
  end for
end for
$\hat{\Omega} = \emptyset$
for $d' = d, d-1, \ldots, 1$ **do**
  for $\omega \in \mathcal{N}$ with $d_\omega = d'$, $U(\{\omega\}) \cap \hat{\Omega} = \emptyset$ **do**
    if $\hat{\sigma}^2_\omega \geq \tau_\omega$ **then** $\hat{\Omega} = \hat{\Omega} \cup \{\omega\}$
    end if
  end for
end for
$\hat{\eta}(x_0) := \frac{1}{2} + \sum_{\omega \in \hat{\Omega} \cup L(\hat{\Omega})} \hat{f}_\omega(x_0)$
Output: $\hat{C}_{\mathrm{HAM}}(x_0) := \mathbb{1}_{\{\hat{\eta}(x_0) \geq 1/2\}}$

d_ω, we do not include a pattern ω in our estimated set $\hat{\Omega}$ whenever $\hat{\sigma}^2_\omega$ is sufficiently small. If, on the other hand, $\hat{\sigma}^2_\omega$ is large then ω is added to the set $\hat{\Omega}$ and we disregard all ω' for which $\omega' \preceq \omega$. This approach ensures that $\hat{\Omega} \in \mathcal{I}(\{0,1\}^d)$. Our final estimate of the regression function η is then based on those $\hat{f}_\omega$ for which the pattern ω is contained in $\hat{\Omega} \cup L(\hat{\Omega})$, that is

$$\hat{\eta}(\cdot) := \frac{1}{2} + \sum_{\omega \in \hat{\Omega} \cup L(\hat{\Omega})} \hat{f}_\omega(\cdot)$$

We classify the test point x_0 according to whether $\hat{\eta}(x_0) \geq 1/2$, or otherwise.

An appealing feature of our algorithm is that we avoid the need for sample-splitting. However, since our proxy for σ^2_ω is calculated by evaluating $\hat{f}_\omega$ at the training data and we condition on the observation indicators in our results, we will require a slightly stronger assumption on the properties of the marginal feature distributions in our class, namely we ask for control of the $\mu_{\omega|o}$ tails. This amounts to a slightly extended version of Definition 3.

DEFINITION 3^+. *Fix $\boldsymbol{\gamma} = (\gamma_1, \ldots, \gamma_d)^T \in [0,\infty)^d$, $C_{\mathrm{L}} \geq 1$ and $\mathcal{O} \subseteq \{0,1\}^d$. Then let $\mathcal{Q}^+_{\mathrm{L}}(\boldsymbol{\gamma}, C_{\mathrm{L}}, \mathcal{O})$ denote the class of distributions for which (9) holds and*

$$\max_{\tilde{o} \in \mathcal{O} : \omega \preceq \tilde{o}} \mu_{\omega|\tilde{o}}\Big(\Big\{x \in \mathbb{R}^d : \min_{o \in \mathcal{O} : \omega \preceq o} \rho_{\mu_{\omega|o}, d_\omega}(x) < \xi\Big\}\Big) \leq C_{\mathrm{L}} \cdot \xi^{\gamma_\omega}$$

for all $\xi > 0$ and all $\omega \in \{0,1\}^d \setminus \{\mathbf{0}_d\}$.

Now, to state our main theoretical result about Algorithm 1, we will write $\mathcal{Q}^+_{\mathrm{Miss}} \equiv \mathcal{Q}^+_{\mathrm{Miss}}(\Omega_\star, c_{\mathrm{E}}, \boldsymbol{\gamma}, C_{\mathrm{L}}, \boldsymbol{\beta}, C_{\mathrm{S}}, \alpha, C_{\mathrm{M}}, \mathcal{O})$ for the class of distributions that arises when we replace $\mathcal{Q}_{\mathrm{L}}(\boldsymbol{\gamma}, C_{\mathrm{L}}, \mathcal{O})$ in the class $\mathcal{Q}'_{\mathrm{Miss}}$ in (11) with $\mathcal{Q}^+_{\mathrm{L}}(\boldsymbol{\gamma}, C_{\mathrm{L}}, \mathcal{O})$.

THEOREM 2. *Fix* $d, n \in \mathbb{N}$, $\mathcal{O} \subseteq \{0,1\}^d$, $o_1, \dots, o_n \in \mathcal{O}$, $\Omega_\star \in \mathcal{I}(\{0,1\}^d) \setminus \{\{\mathbf{0}_d\}, \emptyset\}$, $c_{\mathrm{E}} \in (0, 1/4]$, $\boldsymbol{\gamma} \in [0,\infty)^d$, $C_{\mathrm{L}} \geq 1$, $\boldsymbol{\beta} \in (0,1]^d$, $C_{\mathrm{S}} \geq 1$, $\alpha \in [0,\infty)$ *and* $C_{\mathrm{M}} \geq 1$. *Suppose that* $\Omega_\star \subseteq \mathcal{N}$ *and we have* $\min_{\omega \in \Omega_\star} \gamma_\omega > \max_{\omega \in \Omega_\star} \beta_\omega / d_\omega$. *Suppose that* $n_\omega \geq \log^{\frac{4(\gamma_\omega(2\beta_\omega + d_\omega) + \alpha\beta_\omega)}{\beta_\omega \gamma_\omega}} \big(2|\mathcal{N}| \min_{\omega' \in \Omega_\star} n_{\omega'}\big)$, *for* $\omega \in U(\Omega_\star) \cap \mathcal{N}$, *then there exists a constant* $C_{\mathrm{U}} \geq 1$ *(not depending on* n *nor* $o_1, \dots, o_n$*) such that*

$$\sup_{Q \in \mathcal{Q}^+_{\mathrm{Miss}}} \mathbb{E}_Q \Big\{ \mathcal{E}_{P_Q}(\hat{C}_{\mathrm{HAM}}) \,\Big|\, O_1 = o_1, \dots, O_n = o_n \Big\}$$

$$\leq C_{\mathrm{U}} \cdot \log_+^{\frac{1+\alpha}{2}} \Big(\min_{\omega \in \Omega_\star} n_\omega \Big) \cdot \max_{\omega \in \Omega_\star} \Big\{ n_\omega^{-\frac{\beta_\omega \gamma_\omega (1+\alpha)}{\gamma_\omega(2\beta_\omega + d_\omega) + \alpha\beta_\omega}} \Big\}. \tag{14}$$

The proof of Theorem 2 is presented in Sections 5.3 and S3.2. The key to the proof is to establish in Proposition 6 that $\hat{\Omega} = \Omega_\star$ with high probability, under a sample size condition. The result then follows by combining this with the proof of the upper bound in Theorem 1.

The main conclusion of Theorem 2 is that our classifier $\hat{C}_{\mathrm{HAM}}$ attains the upper bound in Theorem 1, subject to a sample size condition. Note that we do not impose any restrictions on the minimum number of available cases observed for $\omega \in \Omega_\star \cup L(\Omega_\star)$ and that the condition on n_ω for $\omega \in U(\Omega_\star) \cap \mathcal{N}$ only rules out an extreme imbalance between the number of available cases for the different observation patterns. For instance, under our condition on the relationship between the parameters, if we have at least a polylogarithmic proportion of the data available for the patterns in $U(\Omega_\star) \cap \mathcal{N}$, e.g. $n_\omega \geq \log^{8+4(d+\alpha)/\beta_\omega}(2^{d+1} n)$, then the sample size condition holds. The intuition is that we only need a relatively small amount of data to detect that $f_\omega \equiv 0$ for $\omega \in U(\Omega_\star)$.

A thorough investigation of the proof of Theorem 2 in fact allows us to make nontrivial statements about the excess risk of $\hat{C}_{\mathrm{HAM}}$ even when our sample size condition is not satisfied. In particular, we could elucidate the precise consequence of $\hat{\Omega}$ in Algorithm 1 not coinciding with $\Omega_\star$ in small sample settings. First, whenever $\Omega_\star \subseteq \hat{\Omega} \cup L(\hat{\Omega})$, we may seek to control the excess risk in a similar way to (14), with the $\Omega_\star$ in the maximum replaced by $\hat{\Omega}$. On the other hand, if $\Omega_\star \not\subseteq \hat{\Omega} \cup L(\hat{\Omega})$ (or if $\Omega_\star \cap \mathcal{N}^c \neq \emptyset$), then $\hat{C}_{\mathrm{HAM}}$ will estimate a non-zero f_ω by 0, and we will only have a trivial bound on the excess risk.

4. Numerical results. In this section we provide empirical evidence of the strength of the proposed methodology to complement the theoretical results from the previous section. In particular, we will compare the performance of our $\hat{C}_{\mathrm{HAM}}$ classifier applied with canonical values of the class indices $\alpha, \boldsymbol{\beta}, \boldsymbol{\gamma}$, as well as a cross-validation version of our HAM algorithm (see Section S4.1) which only requires a set of possible values that $\alpha, \boldsymbol{\beta}, \boldsymbol{\gamma}$ may take. We also investigate the performance of an oracle version of the HAM classifier which has access to the unknown set $\Omega_\star$. The performance is compared with standard approaches used to deal with missing data in classification problems, namely complete case analysis and several different imputation methods. We also include the comparison with an oracle imputation approach, which has access to the full dataset.

Our experiments[2] will consider the following simulation settings:

Setting 1: We adopt the setting in Example 1. Fix $d = 2$, $u = (2^{1/2}, 0)^T$ and $\Sigma = I$ the 2-dimensional identity matrix. Suppose that $\mathbb{P}(Y = 1) = 1/2$, and $X \mid \{Y = r\} \sim N_d((-1)^r u, \Sigma)$, for $r \in \{0,1\}$ and that $O \perp\!\!\!\perp (X, Y)$. Let $\mathbb{P}(O = \omega) = p^{d_\omega}(1-p)^{d-d_\omega}$,

[2]The code for the simulations is available at https://github.com/TorbenSell/missing_data/.

for $\omega \in \{0,1\}^d$ and homogeneous observation probability $p = 0.7$. In this case, for $x = (x_1, x_2)^T \in \mathbb{R}^2$, the regression function is

$$\eta(x) := \frac{1}{2} + \frac{1}{2}\tanh(-2^{1/2}x_1).$$

It follows that $f_{e_1}(x_1, x_2) = \frac{1}{2}\tanh(-2^{1/2}x_1)$ and that all other f_ω are 0. The corresponding distribution Q of (X, Y, O) belongs to $\mathcal{Q}^+_{\text{Miss}}$, with $\Omega_\star = \{(1,0)^T\}$, any $\boldsymbol{\gamma} = (\gamma_1, 1) \in [0,1) \times \{1\}$, $\alpha = 1$, $\boldsymbol{\beta} = \mathbf{1}_2$, and $\mathcal{O} = \{0,1\}^2$, see Section S1.1 for full details. Therefore, we have

$$R = n_{(1,0)}^{-\frac{2\gamma_1}{3\gamma_1+1}},$$

which can be arbitrarily close to $n_{(1,0)}^{-1/2}$ as $\gamma_1 \to 1$.

Setting 2: In Example 2, take $\mu = U([0,1]^4)$ to be the uniform distribution on the 4-dimensional unit cube. For $x = (x_1, x_2, x_3, x_4)^T \in [0,1]^4$, we let

$$\eta(x) := \frac{1}{2} + \frac{(x_2 - 1/2) \cdot x_3^2}{2} + \frac{x_4 - 1/2}{2}$$

and suppose that $Y \perp\!\!\!\perp O \mid X$. Here the nonzero f_ω functions are

$$f_{e_2}(x) = \frac{(x_2 - 1/2)}{6}; \quad f_{e_2+e_3}(x) = \frac{(x_2 - 1/2)(x_3^2 - 1/3)}{2}; \quad f_{e_4}(x) = \frac{(x_4 - 1/2)}{2}.$$

Here we have $Q \in \mathcal{Q}^+_{\text{Miss}}$, with $\Omega_\star = \{(0,1,1,0)^T, (0,0,0,1)^T\}$, $\boldsymbol{\gamma} = (1, \gamma_2, \gamma_3, \gamma_4)^T$ for any $\gamma_2, \gamma_3, \gamma_4 \in [0, \infty)$, $\alpha = 1$, $\boldsymbol{\beta} = \mathbf{1}_4$, and $\mathcal{O} = \{(1,1,1,0)^T, (1,0,0,1)^T\}$, see Section S1.1. We thus have

$$R = n_{(0,1,1,0)}^{-\frac{2(\gamma_2 \wedge \gamma_3)}{4(\gamma_2 \wedge \gamma_3)+1}} \vee n_{(0,0,0,1)}^{-\frac{2\gamma_4}{3\gamma_4+1}},$$

which, since we can take any arbitrarily large $\boldsymbol{\gamma}$, could be as fast as $R = n_{(0,1,1,0)}^{-1/2} \vee n_{(0,0,0,1)}^{-2/3}$.

Setting 3: We use Example 3, where for $x = (x_1, x_2)^T \in [0,1]^2$ we have

$$\eta(x) := \frac{1}{4} + \frac{x_1}{4} + \frac{1}{4}\cos(4\pi x_2).$$

Here the nonzero f_ω are

$$f_{e_1}(x) = \frac{x_1 - 1/2}{4}; \quad f_{e_2}(x) = \frac{1}{4}\cos(4\pi x_2).$$

We have $Q \in \mathcal{Q}^+_{\text{Miss}}$, with $\Omega_\star = \{e_1, e_2\}$, any $\boldsymbol{\gamma} = (\gamma_1, \gamma_2) \in [0, \infty)^2$, $\alpha = 1$, $\boldsymbol{\beta} = \mathbf{1}_2$ and $\mathcal{O} = \{0,1\}^2$, see Section S1.1. Here, we have

$$R = n_{(1,0)}^{-\frac{2\gamma_1}{3\gamma_1+1}} \vee n_{(0,1)}^{-\frac{2\gamma_2}{3\gamma_2+1}},$$

which approaches $(n_{(1,0)} \wedge n_{(0,1)})^{-2/3}$ as $\gamma_1, \gamma_2 \to \infty$.

In each of these settings, we conduct an experiment by generating a training dataset of size n, for $n \in \{500, 1000\}$. In particular, let $((X_1, Y_1, O_1), \ldots, (X_n, Y_n, O_n)) \sim Q^n$, and a test dataset consisting of $m = 1000$ independent copies of (X, Y), when $(X, Y, O) \sim Q$. For each algorithm, which will only have access to $(X_1^{O_1}, Y_1, O_1), \ldots, (X_n^{O_n}, Y_n, O_n)$, we will present a boxplot of the empirical test error for 100 repeats of the experiment. Here the empirical test error is the proportion of incorrect classifications made on the test dataset. We will compare several different classification algorithms, the first three being our methods:

HAM: Our Algorithm 1 applied with $\boldsymbol{\beta} = \boldsymbol{\gamma} = \mathbf{1}_d$, $\alpha = 1$, and training data $(X_1^{O_1}, Y_1, O_1)$, $\ldots, (X_n^{O_n}, Y_n, O_n)$.

cvHAM: Our Algorithm 1 applied with $\boldsymbol{\beta} = \hat{\boldsymbol{\beta}}$, $\boldsymbol{\gamma} = \hat{\boldsymbol{\gamma}}$, and $\alpha = \hat{\alpha}$, and training data $(X_1^{O_1}, Y_1, O_1), \ldots, (X_n^{O_n}, Y_n, O_n)$. Here $\hat{\boldsymbol{\beta}}$, $\hat{\boldsymbol{\gamma}}$, and $\hat{\alpha}$ are the outputs of our 5-fold cross-validation procedure in Algorithm S1 applied with $A = \{0, 1/2, 1/d\}$, $B = \{1/4, 2/4, 3/4, 1\} \cdot \mathbf{1}_d$ and $G = \{1/2, 1, 2, \infty\} \cdot \mathbf{1}_d$.
Oracle HAM: Our Algorithm 1 with $\boldsymbol{\beta} = \boldsymbol{\gamma} = \mathbf{1}_d$ and $\alpha = 1$, which has access to the true $\Omega_\star$ and thus can omit the estimation of it.

The other methods apply a standard k-nearest neighbour ($k$nn) algorithm (Fix and Hodges, 1952, 1989) to different modifications of the training data, namely

Complete Case (CC): We use only the observations belonging to $N_{\mathbf{1}_d} = \{i \in [n] : O_i = \mathbf{1}_d\}$, and apply the $k$nn algorithm with $k = 1/3 \cdot n_{\mathbf{1}_d}^{-1/(3+d)}$; in other words we apply the $k$nn algorithm directly to $\{(X_i, Y_i) : i \in N_{\mathbf{1}_d}\}$.
Zero Imputation (ZI): All missing values are imputed with the value 0 and we then apply the $k$nn algorithm with $k = 1/3 \cdot n^{-1/(3+d)}$; in other words the $k$nn algorithm is applied directly to $(X_1^{O_1}, Y_1), \ldots, (X_n^{O_n}, Y_n)$.
Mean Imputation (MI): For each $j \in [d]$, let $\hat{\nu}_j := \frac{1}{n_{e_j}} \sum_{i \in N_{e_j}} X_i^{e_j}$ be the mean of the observations in the training data for which the jth variable is not missing and define $\hat{\nu} := \sum_{j \in [d]} \hat{\nu}_j$. Then for each observation $i \in [n]$, let $\tilde{X}_i = X_i^{o_i} + (\mathbf{1}_d - o_i) \odot \hat{\nu}$. We then apply the $k$nn algorithm with $k = 1/3 \cdot n^{-1/(3+d)}$ to $(\tilde{X}_1, Y_1), \ldots, (\tilde{X}_n, Y_n)$.
Multiple Imputation by Chained Equations (MICE): All missing values in the training set are imputed using the 'miceforest' python package to give 5 different datasets. We then apply the $k$nn algorithm with $k = 1/3 \cdot n^{-1/(3+d)}$ to each of these datasets and classify a test point based on a majority vote.
MissForest (MF): All missing values are imputed using the missforest algorithm (Stekhoven and Bühlmann, 2012) which is implemented in the python package 'missingpy'. A $k$nn algorithm with $k = 1/3 \cdot n^{-1/(3+d)}$ is then employed using the imputed dataset as training data.
No missing data (no MD): We apply a $k$nn algorithm with $k = 1/3 \cdot n^{-1/(3+d)}$ to the full training dataset. This can be viewed as an oracle imputation method, which can impute the missing values with the unobserved truth.

The results of our experiments are presented in Figure 2. We see that across our three settings, the HAM classifier performs well, and typically outperforms the widely-used competing methods. In Setting 2, there are no fully observed data points available, so a complete case analysis is not possible. Further we see, especially in Setting 1, that the HAM classifier performs almost as well as its oracle version. In Settings 2 and 3, the slight difference between the performance of HAM and its oracle is due in part to the fact that we have relatively small number of available cases for some observation patterns. For instance, in Setting 3 we need to observe both variables to detect that $f_{\mathbf{1}_2} = 0$ in this example, but on average we only have 188 (when $n = 500$) and 375 (when $n = 1000$) complete cases. Indeed, by inspecting the results, we see that we are slightly conservative in our thresholding step for this setting, nevertheless HAM still performs well. Some further practical improvement may be achievable by tuning the thresholding cutoff depending on the data (i.e. via cross validation).

The results of an additional numerical study with no missing data are presented in Section S4.2 in the online supplementary material, we see that our HAM classifier remains effective in this case.

4.1. *Real data example.* In this section, we illustrate the performance of our HAM algorithm in a real data setting using the 'Mammographic Mass' data set (Elter, 2007) and consider the task of classifying breast cancer tumours as either benign (class 0) or malignant (class 1) based on the shape, margin, and density of the tumour mass, as well as the patient's

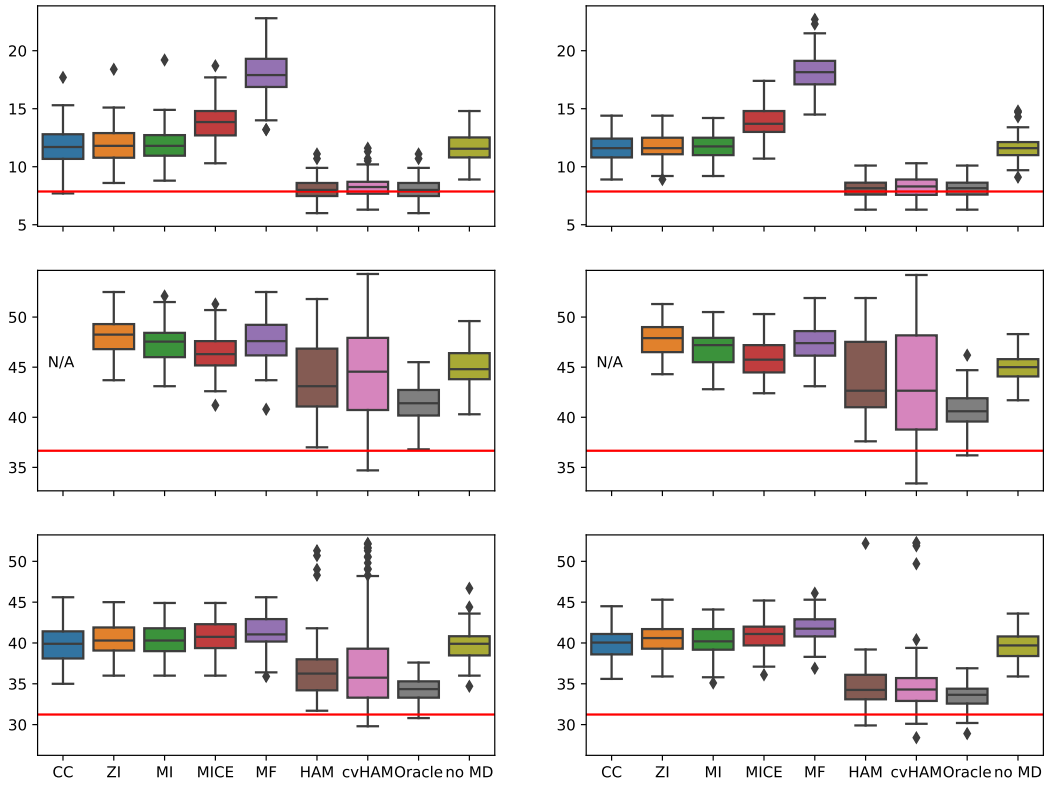


FIG 2. *Box plots of the empirical percentage test errors for the Complete Case (CC), Zero Imputation (ZI), Mean Imputation (MI), MICE, and MissForest (MF) approaches and our classifier with known parameters (HAM) and our classifier with parameters estimated by a cross-validation procedure (cvHAM). For reference, we also include our Oracle HAM (Oracle) classifier, and a classifier that has access to all data, with no missing values (no MD). The Bayes risk is shown as the red horizontal line. We present the results for Setting 1 (top row), Setting 2 (middle row) and Setting 3 (bottom row), with* $n = 500$ *(left column) and* $n = 1000$ *(right column).*

age. There are 961 observations in the dataset, of which 5 have missing entries for age, 31 have no entry for mass shape, 48 are missing entries for mass margin, and 76 have no entry for mass density. There are 516 benign tumours (53.7%) and 445 malignant tumours (46.3%).

We compare our HAM and cvHAM algorithms to the complete case classifier (CC), as well as Zero Imputation (ZI), Mean Imputation (MI), MICE, and MissForest (MF), all implemented as described in the previous section. We randomly split the data into a training set and test set by sampling 161 of the complete cases to form our test set, leaving $n = 800$ observations to form the training set which contains missing values. We record the test error of each algorithm over 100 different train-test splits. The results of this experiment are shown in Figure 3, where we see that our HAM and cvHAM approaches are outperforming the existing methods.

5. Proofs.

5.1. *Constructions used in the lower bound proof of Theorem 1.* Our proof of the lower bound in Theorem 1 makes use of a new version of Assouad's Lemma, modified to our classification with missing data setting (see Lemma S5 in Section S2). To illustrate the key ideas of our proof, we initially consider the case that $\Omega_\star = \{\omega\}$ contains only one element. We construct a finite set of distributions Q of (X, Y, O) belonging to $\mathcal{Q}'_{\text{Miss}}$. The full constructions

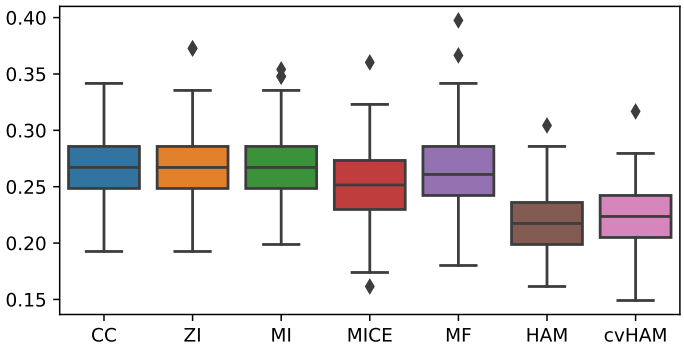


FIG 3. *Box plots of the empirical percentage test errors for the Complete Case (CC), Zero Imputation (ZI), Mean Imputation (MI), MICE, and MissForest (MF) approaches and our classifier with known parameters (HAM) and our classifier with parameters estimated by a cross-validation procedure (cvHAM).*

for our proofs are somewhat complicated and many of the details, as well as the corresponding theory and ultimately the full proof of the lower bound in Theorem 1 for a general $\Omega_\star$, are presented in Section S2.

Here we present some of the key aspects of our proof. First, for each distribution Q that we construct, we take (X, Y) and O to be independent and take the distribution of O to be uniform on $\mathcal{O}$, so $Q \in \mathcal{Q}_{\text{Miss}}(\mathcal{O})$. In order to ensure that our regression functions decompose as in (6), we require a carefully designed marginal feature distribution, which in particular satisfies many symmetries about the origin. The full definition of these, which differ slightly in the *light tailed* case (where $\gamma_\omega \geq 1$) and the *heavy tailed* case (where $\gamma_\omega < 1$), are given in Sections 5.1.1 and 5.1.2. In both cases, we have a discrete lattice of support points on which the regression function construction will change between distributions, and a set where the regression functions does not change between distributions in our construction. Sections S2.2 and S2.5 in the supplementary material show that the distributions are constructed in such as way that they belong to the class $\mathcal{Q}^+_{\text{L}}(\boldsymbol{\gamma}, C_{\text{L}}, \mathcal{O})$.

We then, in Sections S2.3 and S2.6, define the regression function η for our respective light- and heavy-tailed initial constructions when $\Omega_\star = \{\omega\}$. The symmetry in marginal distributions allows us to define η via a single nonzero function f_ω in the anova decomposition in (6). This function f_ω is defined first on the region $(0, \infty)^d$ and then mirrored or flipped on the appropriate axes to define the function on the whole of $\mathbb{R}^d$, ensuring that $\eta = 1/2 + f_\omega$, i.e. (6) holds with all other $f_{\omega'} = 0$ (see Lemmas S12 and S19. On the lattice, f_ω is taken to be either ϵ or $-\epsilon$ for some small $\epsilon > 0$. The different possible combinations of $\pm\epsilon$ on these points lead to the different distributions used in our application of Assouad's Lemma. On the remainder of the marginal support, f_ω is taken to be a particular β-smooth function. This choice allows us to extend the definition of f_ω to a β-smooth function on the whole of $\mathbb{R}^d$ (via McShane's extension Theorem – see Corollaries S11 and S18). Lemmas S12 and S19 also show that the corresponding distributions belong to $\mathcal{P}_{\text{S}}(\boldsymbol{\beta}, 1)$. Concluding Sections S2.3 and S2.6, Lemmas S13 and S20 show that the distributions belong to $\mathcal{P}_{\text{M}}(\alpha, C_{\text{M}})$.

Sections S2.4 and S2.7 complete the proof of the lower bound when $\Omega_\star = \{\omega\}$ for the light- and heavy-tailed cases, respectively. More precisely, we apply our version of Assouad's Lemma to carefully selected versions of the constructions described above. The full proof of the lower bound in Theorem 1 is presented in Section S2.8. The proof extends the constructions above to the general $\Omega_\star$ case, using a mixture of distributions, which establishes the lower bound over $\mathcal{Q}'_{\text{Miss}}(\Omega_\star, c_{\text{E}}, \boldsymbol{\gamma}, C_{\text{L}}, \boldsymbol{\beta}, C_{\text{S}}, \alpha, C_{\text{M}}, \mathcal{O})$; in fact we will see that our constructions belong to the smaller class $\mathcal{Q}^+_{\text{Miss}}$, and thus that our lower bound holds even over this more restrictive class.

5.1.1. *The marginal feature distribution in the light tailed case.* It is convenient to introduce some additional notation at this point. For a set $A^+ \subseteq [0,\infty)^d$ and $s = (s_1,\ldots,s_d)^T \in \{-1,1\}^d$, we will write $s \odot A^+ := \{x = (x_1,\ldots,x_d)^T \in \mathbb{R}^d : x = s \odot x^+, x^+ \in A^+\} \subseteq \mathbb{R}^d$. We now define the marginal distribution of the feature vectors, which will in fact be the same for all Q in our class. First, for $q \in \mathbb{N}$ and $r > 0$, let

$$\mathcal{T}_{q,r} = \left\{ \left(1 + \frac{rv_1}{q}, \ldots, 1 + \frac{rv_d}{q}\right)^T : v_j \in [q] \text{ for } j \in [d] \right\}.$$

Then, fixing $\omega \in \{0,1\}^d \setminus \{\mathbf{0}_d\}$, let $\mathcal{T}^\omega_{q,r} = \{z^\omega \in \mathbb{R}^d : z \in \mathcal{T}_{q,r}\}$, $T := \left|\mathcal{T}^\omega_{q,r}\right| = q^d$ and enumerate this set as $\{z^\omega_1,\ldots,z^\omega_T\} \subseteq [0,1+r]^d$. Further let $\Pi_\omega : \mathbb{R}^d \to \mathbb{R}^{d_\omega}$ be the projection of $\mathbb{R}^d$ onto the d_ω non-zero coordinates of ω, i.e. for $x := (x_1,\ldots,x_d)^T \in \mathbb{R}^d$ and $\omega := (\omega_1,\ldots,\omega_d)^T \in \{0,1\}^d$, we have $\Pi_\omega(x) := (x_j)_{j\in[d]:\omega_j=1} \in \mathbb{R}^{d_\omega}$. For $A \subseteq \mathbb{R}^d$, let $\Pi_\omega(A) := \{\Pi_\omega(x) : x \in A\}$. Further, for $\kappa > 0$, let $\mathcal{R} := [1+2r, 1+2r+\kappa]^d$ and $\mathcal{R}^\omega := \{x^\omega \in \mathbb{R}^d : x \in \mathcal{R}\}$. Let $\mathcal{R}_0 := [1,1+r]^d$, $\mathcal{R}^\omega_0 := \{x^\omega \in \mathbb{R}^d : x \in \mathcal{R}_0\}$ and, for $j \in \{\tilde{j} \in [d] : \omega_{\tilde{j}} = 1\}$, let $\mathcal{R}^\omega_j = \mathcal{R}^\omega_0 + (1+2r+\kappa)\cdot e_j$.

We define the marginal X distribution as follows: for $a,b \in [0,1/2]$ and $A \subseteq \mathbb{R}^d$ (measurable), let

$$\begin{aligned}\mu(A) \equiv \mu^{(\omega)}_{\kappa,r,q,a,b}(A) := {} & \frac{a}{2^d T} \sum_{s\in\{-1,1\}^d} \sum_{t=1}^{T} \mathbb{1}_{\{s^\omega \odot z^\omega_t \in A\}} \\ & + \frac{b}{2^d d_\omega r^{d_\omega}} \sum_{j\in[d]:\omega_j=1} \sum_{s\in\{-1,1\}^d} \mathcal{L}_{d_\omega}\Big(\Pi_\omega\big(A \cap (s \odot \mathcal{R}^\omega_j)\big)\Big) \\ & + \frac{1-a-b}{2^d \kappa^{d_\omega}} \sum_{s\in\{-1,1\}^d} \mathcal{L}_{d_\omega}\Big(\Pi_\omega\big(A \cap (s \odot \mathcal{R}^\omega)\big)\Big).\end{aligned} \tag{15}$$

A particular instance of the support of this construction is given in Figure 4. The regression functions in this case are given in Section S2.3. The main idea is that we will take $\eta = 1/2 + f_\omega$ to be slightly above or below $1/2$ on each of the grid points in $\mathcal{T}^\omega_{q,r}$, and bounded away from $1/2$ on the remainder of the support. This is initially defined only on the "top right" quadrant of the support and then reflected asymmetrically to ensure the f_ω satisfies the orthogonality constraint in the anova decomposition – see (S7) in Section S2.3.

Note that the construction of the marginal distribution used in this subsection shares some aspects with the construction used by Reeve et al. (2021) in their transfer learning problem. In particular, the regions $\mathcal{T}^\omega_{q,r}$ and $\mathcal{R}^\omega$ are used similarly to how they were used in Reeve et al. (2021), while here we introduce the sets $\mathcal{R}^\omega_j$ for $j \in [d]$ and carefully *reflect* the upper-right quadrant in the coordinate axes to give our final X marginal construction. These latter aspects are crucial for constructing distributions Q satisfying Definitions 1 to 5 in our missing data problem.

5.1.2. *The marginal feature distribution in the heavy tailed case.* The marginal construction for the heavy tailed case is somewhat simpler, in particular the marginal distribution of X can be taken to be a product measure. Let ν_0 denote a point mass at zero, i.e. $\nu_0(A) = \mathbb{1}_{\{0\in A\}}$, for $A \subseteq \mathbb{R}$. For $\tilde{q} \in \mathbb{N}$, let $\nu_1 \equiv \nu_{1,\tilde{q}}$ denote the distribution on $\mathbb{R}$ given by

$$\nu_{1,\tilde{q}}(A) := \frac{1}{2\tilde{q}} \sum_{j=1}^{\tilde{q}} \mathbb{1}_{\{1+j/\tilde{q}\in A\}} + \mathbb{1}_{\{-1-j/\tilde{q}\in A\}},$$

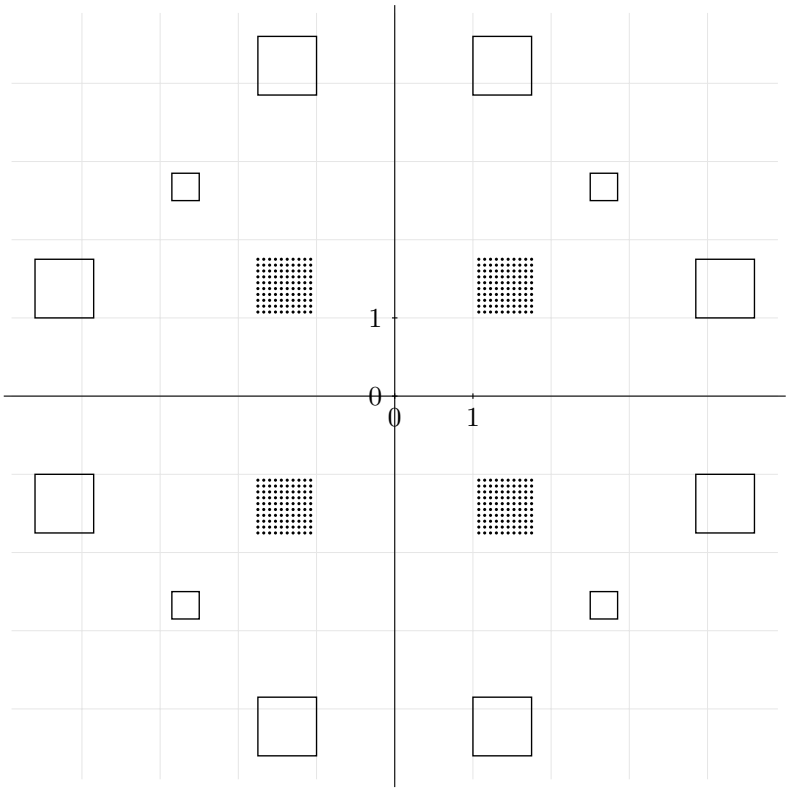


FIG 4. *The support of the light tailed marginal for an observation pattern ω with $d_\omega = 2$ and $\gamma_\omega \geq 1$. The lattice in the top right quadrant is $\mathcal{T}^\omega_{q,r}$, the boxes are $\mathcal{R}^\omega$, $\mathcal{R}^\omega_1$ and $\mathcal{R}^\omega_2$. The boxes carry enough mass to ensure that projecting onto one variable results in light tails. Specifically here we plot the support of $\mu^{(\mathbf{1}_2)}_{\kappa,r,q,a,b}$ with $\kappa = 0.35$, $r = 0.75$ and $q = 10$.*

for $A \subseteq \mathbb{R}$. Further, for $a \in [0,1], r > 1$ and $q \in \mathbb{N}$, let $\nu_2 \equiv \nu_{2,q,r,a}$ denote the distribution on $\mathbb{R}$ given by

$$\nu_{2,q,r,a}(A) := \frac{a}{2q}\sum_{j=1}^{q}\Big(\mathbb{1}_{\{1+\frac{rj}{q}\in A\}} + \mathbb{1}_{\{-1-\frac{rj}{q}\in A\}}\Big) + \frac{1-a}{2}\mathcal{L}_1\Big(A \cap \big\{(-r-2,-r-1)\cup(r+1,r+2)\big\}\Big).$$

Now fix $\omega \in \{0,1\}^d \setminus \{\mathbf{0}_d\}$ and $\tilde{j} \in \{j \in [d] : \omega_j = 1\}$. Let

$$\mu \equiv \mu^{(\omega)}_{q,\tilde{q},r,a,\tilde{j}} = \bigotimes_{j=1}^{\tilde{j}-1}\nu_{\omega_j} \otimes \nu_{2,q,r,a} \otimes \bigotimes_{j=\tilde{j}+1}^{d}\nu_{\omega_j}. \tag{16}$$

A particular instance of the support of this construction is given in Figure 5. The corresponding regression function in this case is conceptually similar to that in the light tailed case above and is therefore deferred to Section S2.6.

5.2. *Outline of the proof of the upper bound in Theorem 1.* In this section, we provide the proof of the upper bound in Theorem 1. Our strategy for the proof is to introduce two high-probability events $E^\delta_1(x)$ and $E^\delta_2(x)$. The first provides control of the nearest neighbour

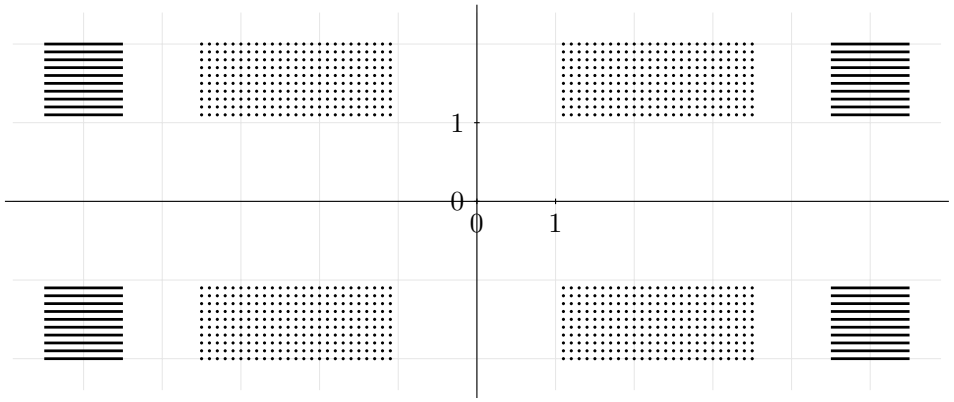


FIG 5. *The support of the heavy tailed marginal for an observation pattern ω with $d_\omega = 2$ and $\gamma_\omega < 1$. Specifically here we plot the support of $\mu^{(\mathbf{1}_2)}_{q,\tilde{q},r,a,1} = \nu_{2,q,r,a} \otimes \nu_{1,\tilde{q}}$, with $q = 25$, $\tilde{q} = 10$, $r = 2.5$. The lines on the sides ensure that projecting onto the x_2 axis results in light tails.*

distances to a point x in the support of μ for each observation pattern $\omega \in \mathcal{N}$. Our second high-probability event asks for control of the empirical average of the labels of the nearest neighbours in each observation pattern. Then, working on these events, we provide in Lemma 3 control of the error $|\hat{f}_\omega(x) - f_\omega(x)|$ for each $\omega \in \mathcal{N}$. This establishes the rate at which we are able to estimate the functions f_ω for $\omega \in \mathcal{N}$.

We next show that in regions where η is sufficiently far from a half, a version of our classifier agrees with the Bayes classifier with high probability. The main result in this subsection is Proposition 5, which provides a tail bound on the excess risk of this version of our algorithm. We then prove the upper bound in Theorem 1 at the end of this subsection, by integrating the tail bound for an oracle version of the algorithm, which has access to the true $\Omega_\star$. The proofs of the results in subsection, as well as some additional lemmas are given in Section S3.1 in the online supplement.

Before presenting the results, it is convenient to fix some notation that we will use throughout this section. First, recall from the statement of Theorem 1 that we are treating the missingness indicators $o_1, \ldots, o_n \in \mathcal{O} \subseteq \{0,1\}^d$ as fixed and all probability statements in this section should be interpreted as being conditional on $O_1 = o_1, \ldots, O_n = o_n$. For $\omega \in \{0,1\}^d$, we will write $\rho_\omega(\cdot)$ in place of $\min_{o \in \mathcal{O}: \omega \preceq o}\{\rho_{\mu_{\omega|o}, d_\omega}(\cdot)\}$. Let $k_{\mathbf{0}_d} = n$ and recall from Algorithm 1 that $k_\omega = 1 + \lfloor n_\omega^{\frac{2\beta_\omega\gamma_\omega}{\gamma_\omega(2\beta_\omega + d_\omega) + \alpha\beta_\omega}} \rfloor$, for $\omega \in \mathcal{N} \setminus \{\mathbf{0}\}$. Let $\mathcal{X} := \{x \in \mathbb{R}^d : \rho_\omega(x^\omega) > 0, \text{ for all } \omega \in \mathcal{N}\}$. For $\delta \in (0,1]$, $\omega \in \mathcal{N}$, $k_\omega \in [n_\omega]$, and $x \in \mathcal{X}$, let $\tilde{k}_\omega \equiv \tilde{k}_\omega(x) := \lceil 4\log_+(|\mathcal{N}|/\delta) \rceil \mathbb{1}_{\{k_\omega < \lceil 4\log_+(|\mathcal{N}|/\delta)\rceil\}} + k_\omega \mathbb{1}_{\{\lceil 4\log_+(|\mathcal{N}|/\delta)\rceil \le k_\omega < n_\omega\rho_\omega(x^\omega)/2\}} + (\lceil n_\omega\rho_\omega(x^\omega)/2 \rceil - 1)\mathbb{1}_{\{n_\omega\rho_\omega(x^\omega)/2 \le k_\omega\}}$ if $\lceil 4\log_+(|\mathcal{N}|/\delta)\rceil < n_\omega\rho_\omega(x^\omega)/2$, and let $\tilde{k}_\omega \equiv \tilde{k}_\omega(x) := 0$ if $\lceil 4\log_+(|\mathcal{N}|/\delta)\rceil \ge n_\omega\rho_\omega(x^\omega)/2$.

We now formally introduce our two high-probability events. First, for $x \in \mathcal{X}$ and $\omega \in \mathcal{N}$, write $X^\omega_{(0)_\omega}(x^\omega) = x^\omega$ and define the event

$$A^\delta_\omega(x) := \left\{ \|X^\omega_{(\tilde{k}_\omega)_\omega}(x^\omega) - x^\omega\| \le \left(\frac{2\tilde{k}_\omega}{n_\omega\rho_\omega(x^\omega)}\right)^{1/d_\omega} \right\}.$$

Let $E^\delta_1(x) := \bigcap_{\omega \in \mathcal{N}} A^\delta_\omega(x)$. Lemma S25 in Section S3.1 bounds the probability of the event $E^\delta_1(x)$. We now define the high-probability event on which the empirical average of labels of the nearest neighbours is close to the corresponding average of the regression function evaluated at the feature vectors. For $\omega \in \{0,1\}^d$, write $\eta_\omega : \mathbb{R}^d \to [0,1]$ for the regression

function of Y given X^ω given by $\eta_\omega(x) := \mathbb{P}_Q(Y = 1 \mid X^\omega = x^\omega)$. For $\delta \in (0,1)$, $\omega \in \mathcal{N}$ and $x \in \mathbb{R}^d$ let

$$B^\delta_\omega(x) := \Big\{ \Big| \frac{1}{k_\omega} \sum_{i=1}^{k_\omega} \{ Y_{(i)_\omega}(x) - \eta_\omega(X_{(i)_\omega}(x)) \} \Big| \leq \sqrt{\frac{\log_+(2|\mathcal{N}|/\delta)}{2k_\omega}} \Big\},$$

and let $E^\delta_2(x) := \bigcap_{\omega \in \mathcal{N}} B^\delta_\omega(x)$. Lemma S26 in Section S3.1 bounds the probability of this event.

Our first result here controls the bias and variance of $\hat{f}_\omega$ for $\omega \in \mathcal{N}$ on the event $E^\delta_1(x) \cap E^\delta_2(x)$. To this end, we will make use of a universal bound on $\|f_\omega\|_\infty$, given by C_{B} in Proposition S4. For $\omega \in \mathcal{N} \setminus \{\mathbf{0}\}$ and $x \in \mathcal{X}$, let

$$R_{\omega,1}(x) = 2C_{\mathrm{B}}C_{\mathrm{S}} \cdot \max_{\omega' \preceq \omega} \Big(\frac{2\tilde{k}_{\omega'}}{n_{\omega'} \rho_{\omega'}(x^{\omega'})} \Big)^{\beta_{\omega'}/d_{\omega'}}; \quad R_{\omega,2} := \sqrt{\frac{\log_+(2|\mathcal{N}|/\delta)}{2k_\omega}}, \tag{17}$$

and let $R_{\mathbf{0}_d,1}(x) = 0$ and $R_{\mathbf{0}_d,2} = \sqrt{\frac{\log_+(2|\mathcal{N}|/\delta)}{2n}}$.

LEMMA 3. *Fix $\omega \in \mathcal{N}$ and $x \in \mathcal{X}$, then on the event $E^\delta_1(x) \cap E^\delta_2(x)$ we have that*

$$\big| \hat{f}_\omega(x) - f_\omega(x) \big| \leq \kappa_\omega \cdot \big(2^{d_\omega} \cdot R_{\omega,1}(x) + R_{\omega,2} \big), \tag{18}$$

for some $1 \leq \kappa_\omega \leq 2(d_\omega + 1)^{d_\omega}$, which is given explicitly in the proof.

It is now convenient to define a version of our HAM classifier that depends on a deterministic set $\Omega \subseteq \mathcal{N}$. Indeed, let $\hat{\eta}_\Omega(x_0) := 1/2 + \sum_{\omega \in \Omega \cup L(\Omega)} \hat{f}_\omega(x_0)$ and let $\hat{C}_\Omega(x_0) := \mathbb{1}_{\{\hat{\eta}_\Omega(x_0) \geq 1/2\}}$. Our next two results concern the properties of $\hat{C}_\Omega(x_0)$, and we begin by establishing the rate at which $\hat{\eta}_\Omega(x)$ approximates $\eta(x)$ on the event $E^\delta_1(x) \cap E^\delta_2(x)$. For $x \in \mathcal{X}$ let $R^\delta_{\mathbf{0}_d}(x) := \sqrt{\frac{\log_+(2|\mathcal{N}|/\delta)}{2n}}$ and for $\omega \in \mathcal{N} \setminus \{\mathbf{0}_d\}$ and $x \in \mathcal{X}$ define

$$R^\delta_\omega(x) := \left(\frac{n_\omega^{-\frac{\alpha\beta^2_\omega/d_\omega}{\gamma_\omega(2\beta_\omega + d_\omega) + \alpha\beta_\omega}}}{\rho^{\beta_\omega/d_\omega}_\omega(x^\omega)} + \log^{1/2}_+(2|\mathcal{N}|/\delta) \right) n_\omega^{-\frac{\beta_\omega\gamma_\omega}{\gamma_\omega(2\beta_\omega + d_\omega) + \alpha\beta_\omega}}.$$

This represents the rate at which $\hat{f}_\omega(x)$ estimates $f_\omega(x)$ on our high-probability event. Further, for $\Omega \in \{0,1\}^d$ satisfying $\Omega_\star \subseteq \Omega \cup L(\Omega)$ and $x \in \mathcal{X}$, let

$$R^\delta_\Omega(x) := \max_{\omega \in \Omega} R^\delta_\omega(x).$$

Lemma 4 shows that $\hat{C}_\Omega$ agrees with C^{Bayes} on the event $E^\delta_1(x) \cap E^\delta_2(x)$, whenever η is not too close to $1/2$.

LEMMA 4. *Fix $d, n \in \mathbb{N}$, $\mathcal{O} \subseteq \{0,1\}^d$, $o_1, \ldots, o_n \in \mathcal{O}$, $\Omega_\star, \Omega \in \mathcal{I}(\{0,1\}^d) \setminus \{\{\mathbf{0}_d\}, \emptyset\}$, $c_{\mathrm{E}} \in (0,1)$, $\boldsymbol{\gamma} \in [0,\infty)^d$, $C_{\mathrm{L}} \geq 1$, $\boldsymbol{\beta} \in (0,1]^d$, $C_{\mathrm{S}} \geq 1$, $\alpha \in [0,\infty)$, $C_{\mathrm{M}} \geq 1$ and $\delta \in (0,1)$. Suppose that $\Omega_\star \subseteq \Omega \cup L(\Omega) \subseteq \mathcal{N}$. Fix $Q \in \mathcal{Q}'_{\mathrm{Miss}}$. Let*

$$C := \max_{\omega \in \Omega} \Big\{ 2\kappa_\omega |\Omega \cup L(\Omega)| \big(2^{3+d_\omega} C_{\mathrm{B}} C_{\mathrm{S}} + \sqrt{1/2} \big) \Big\}.$$

If $x \in \mathcal{X}$ is such that

$$|\eta(x) - 1/2| \geq C \cdot R^\delta_\Omega(x),$$

then, on the event $E^\delta_1(x) \cap E^\delta_2(x)$, we have $\hat{C}_\Omega(x) = C^{\mathrm{Bayes}}(x)$.

We are now in a position to state Proposition 5.

PROPOSITION 5. *Fix* $d, n \in \mathbb{N}$, $\mathcal{O} \subseteq \{0,1\}^d$, $o_1, \ldots, o_n \in \mathcal{O}$, $\Omega_\star, \Omega \in \mathcal{I}(\{0,1\}^d) \setminus \{\{\mathbf{0}_d\}, \emptyset\}$, $c_{\mathrm{E}} \in (0,1)$, $\boldsymbol{\gamma} \in [0,\infty)^d$, $C_{\mathrm{L}} \geq 1$, $\boldsymbol{\beta} \in (0,1]^d$, $C_{\mathrm{S}} \geq 1$, $\alpha \in [0,\infty)$, $C_{\mathrm{M}} \geq 1$ *and* $\delta \in (0,1)$. *Let* $\delta' := \frac{\delta}{2} \max_{\omega \in \Omega}\{1/n_\omega^{1+\alpha}\}$. *Suppose that* $\Omega_\star \subseteq \Omega \cup L(\Omega) \subseteq \mathcal{N}$ *and* $\min_{\omega \in \Omega} \gamma_\omega > \max_{\omega \in \Omega} \beta_\omega / d_\omega$. *There exists a constant* $K_\Omega \geq 2$ *such that, for any* $Q \in \mathcal{Q}'_{\mathrm{Miss}}$, *we have*

(19)
$$\mathbb{P}_Q\left\{\mathcal{E}_P(\hat{C}_\Omega) > K_\Omega \cdot \log_+^{\frac{1+\alpha}{2}}(2|\mathcal{N}|/\delta') \cdot \max_{\omega \in \Omega} n_\omega^{-\frac{\beta_\omega \gamma_\omega (1+\alpha)}{\gamma_\omega(2\beta_\omega + d_\omega) + \alpha\beta_\omega}} \,\middle|\, O_1 = o_1, \ldots, O_n = o_n\right\} \leq \delta.$$

The idea behind the proof of this result is to integrate the bound on the regression function in Lemma 4 over $\mathcal{X}$. This involves ruling out an exceptional set on which the event $E_1^{\delta'}(x) \cap E_2^{\delta'}(x)$ fails to hold; see Lemma S28 in Section S3.1 for full details. The proof of the upper bound in Theorem 1 now follows by bounding the expectation of the excess risk using (19), and replacing Ω with $\Omega_\star$. The details are also deferred to Section S3.1.

5.3. *Outline of the proof of Theorem 2.* In this subsection we provide an outline for the proof of Theorem 2. The full details of the proof are somewhat lengthy and are therefore given in Section S3.2 in the online supplementary material. The key idea is to show that our estimator $\hat{\Omega}$ used in the HAM classifier in Algorithm 1 correctly identifies $\Omega_\star$ with high probability; see Proposition 6 below. The proof of Theorem 2 then follows by carefully combining this proposition with the upper bound in Theorem 1.

Before stating the proposition, we will use the constant C_{B} (which provides an upper bound on $\|f_\omega\|_\infty$ and only depends on d) from Proposition S4, and define $\phi_\omega := \gamma_\omega(2\beta_\omega + d_\omega) + \alpha\beta_\omega$, $\kappa_{\omega,\circ} := 2\phi_\omega/(\beta_\omega\gamma_\omega)$ and $C_\rho := (32 C_{\mathrm{B}}^2 C_{\mathrm{L}})^{-\frac{1}{\min_{\omega \in \mathcal{N}} \gamma_\omega}}$. Further, we will make use of a constant $\kappa_{\omega,\mathrm{T}}$, which only depends on d_ω; see the proof of Lemma S34 in Section S3.2 for a precise definition.

PROPOSITION 6. *Fix* $d, n \in \mathbb{N}$, $\mathcal{O} \subseteq \{0,1\}^d$, $o_1, \ldots, o_n \in \mathcal{O}$, $\Omega_\star \in \mathcal{I}(\{0,1\}^d) \setminus \{\{\mathbf{0}_d\}, \emptyset\}$, $c_{\mathrm{E}} \in (0, 1/4]$, $\boldsymbol{\gamma} \in [0,\infty)^d$, $C_{\mathrm{L}} \geq 1$, $\boldsymbol{\beta} \in (0,1]^d$, $C_{\mathrm{S}} \geq 1$, $\alpha \in [0,\infty)$, $C_{\mathrm{M}} \geq 1$ *and* $\delta \in (0,1)$. *Suppose that* $\Omega_\star \subseteq \mathcal{N}$ *and* $\min_{\omega \in \Omega_\star} \gamma_\omega > \max_{\omega \in \Omega_\star} \beta_\omega / d_\omega$. *Suppose further that* $(n_\omega)_{\omega \in \mathcal{N}}$ *satisfies*

$$n_\omega \geq \left(\frac{2^7 C_B^2 C_{\mathrm{L}}}{c_{\mathrm{E}}}\right)^{\kappa_{\omega,\circ}(1 \vee \frac{1}{2\alpha})} \left(\frac{20 C_{\mathrm{S}} \kappa_{\omega,\mathrm{T}} \kappa_{\omega,\circ} \cdot \log_+(2|\mathcal{N}|n_\omega/\delta)}{C_\rho^{1 \vee \gamma_\omega}}\right)^{2\kappa_{\omega,\circ}}$$

for all $\omega \in \Omega_\star$, *and*

$$n_\omega \geq \left(16 \kappa_{\omega,\circ} \kappa_{\omega,T}^2 \cdot \log(2|\mathcal{N}|n_\omega/\delta)\right)^{\kappa_{\omega,\circ}},$$

for all $\omega \in \mathcal{N} \cap U(\Omega_\star)$. *Then we have*

$$\sup_{Q \in \mathcal{Q}^+_{\mathrm{Miss}}} \mathbb{P}_Q\left(\hat{\Omega} \neq \Omega_\star \,\middle|\, O_1 = o_1, \ldots, O_n = o_n\right) \leq 4\delta.$$

Proposition 6 shows that, under a sample size condition, we have $\hat{\Omega} = \Omega_\star$ with high probability, uniformly over the class $\mathcal{Q}^+_{\mathrm{Miss}}$. For the proof of this result, given in Section S3.2, we introduce several additional high-probability events, which control how well we can estimate σ_ω^2 as well as the error of $\hat{f}$ at each of the training data points simultaneously. On the intersection of these events, we are then able to show that $\hat{\sigma}_\omega^2$ is on the correct side of the threshold τ_ω for each relevant $\omega \in \mathcal{N}$. The sample size condition ensures that we have enough signal in our training data to detect that f_ω is nonzero. Note, however, that in Theorem 2 we are in fact able to remove the condition on n_ω for $\omega \in \Omega_\star$ and simplify the condition for $\omega \in \mathcal{N} \cap U(\Omega_\star)$.

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